Algebraic Topology Class Notes

These are class notes for a first year graduate level Algebraic Topology class. They draw mostly from Allen Hatcher's 'Algebraic Topology (2002)' [Ha] which is available on his webpage at 'http://www.math.cornell.edu/ hatcher/'.

PART I : Homotopy

1 Definition of the Fundamental Group

The basic idea of the fundamental group of a space Y is that it counts the ways we can embed the circle S^1 into Y. If we can continuously deform the one embedding to another, then we should consider them the same embedding. The idea of deforming embedding of a space X into a space Y is made formal with *homotopies*.

1.1 Paths

Definition 1.1. A *path* in a space X is a continuous function $f : I \to X$. Where $x_0 = f(0)$ and $x_1 = f(1)$ for a path f in X, we say f is a path *from* x_0 *to* x_1, and we call x_0 and x_1 the *endpoints* of f.

Working towards a 'group of paths' in a space X we define the *inverse* $f^{-1} : I \to X$ of a path f from x_0 to x_1 to be the path from x_1 to x_0 traversing f in the other direction:

$$f^{-1}(i) = f(1-i)$$

If f is a path from x_0 to x_1 and g is a path from x_1 to x_2 then the product (or composition) of f and g is the path $f \cdot g : I \to X$ defined:

$$f \cdot g(i) = \begin{cases} f(2i) & \text{if } i \leq 1/2 \\ g(2i-1) & \text{if } i > 1/2 \end{cases}$$

Problem 1.1. Show that $f \cdot g$ is indeed a path. That is, that the function defined is continuous.

The product, of course, is not defined for every pair of paths, so it is not really a product. Nor do we have a multipicative identity, so we have some work to do.

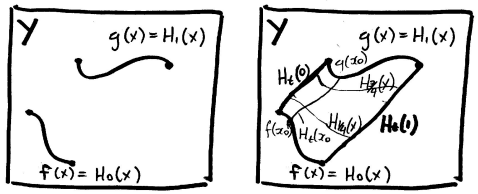


Figure 1: Left: embeddings f and g of $X = I$ into Y. Right: a homotopy H between f and g.

1.2 Homotopy

Let $I = [0,1]$ be the unit interval, a closed subspace of $\mathbb{R}$. Let X and Y be topological spaces, and f and g be continous functions from X to Y. A *homotopy* between f and g is a family $\{H_t\}_I$ of functions $H_t : X \to Y$ such that

i. $H_0 = f$ and $H_1 = g$,

ii. each H_t is continuous, and

iii. for each $x_0 \in X$, the function $t \mapsto H_t(x_0)$ is a path in Y.

We use t for an index, as we view the homotopy as f deforming to g over time 0 to 1. The point $f(x_0)$ is seen as moving along the path $t \mapsto H_t(x_0)$. Lets define homotopy formally.

Definition 1.2. A *homotopy* H between continuous functions $f, g : X \to Y$ is a continuous function $H : X \times I \to Y$ such that $H(x,0) = f(x)$ and $H(x,1) = g(x)$ for all $x \in X$. Two functions f and g are *homotopic* if there is a homotopy between them. A function is *null-homotopic* if it is homotopic to a constant map.

A homotopy defines a family $\{H_t\}_I$ of functions $H_t : X \to Y$ by $H_t(x) = H(x,t)$. The notation $H(\cdot, t) = H_t$ and $H(x, \cdot) : I \to Y : t \mapsto H(x,t)$ will also be useful.

Problem 1.2. Show that this family $\{H_t\}_I$ has the properties ii. and iii. given above.

Given a homotopy $H^a : X \times I \to Y$ between $H^a_0 = f$ and $H^a_1 = g$ and $H^b : X \times I \to Y$ between $H^b_0 = g$ and $H^b_1 = h$, the function $H^a \cdot H^b : X \times I \to Y$ defined by

$$(H^a \cdot H^b)(x, \cdot) = H^a(x, \cdot) \cdot H^b(x, \cdot)$$

is a homotopy between f and g. So the relation of being homotopic is transitive.

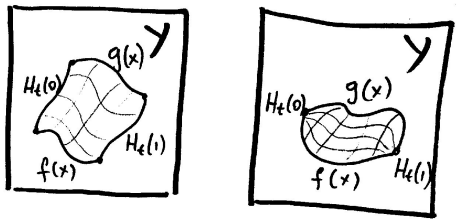


Figure 2: Left: typical homotopy of embeddings of $X = I$. Right: typical homotopy of paths.

Problem 1.3. Show that the relation of being homotopic is an equivalence relation.

Problem 1.4. Let $H^a : X \times I \to Y$ be a homotopy between f^a and g^a, and $H^b : Y \times I \to Z$ be a homotopy between f^b and g^b. Show that $H^b \circ H^a : X \times I \to Z$ defined by $H^b \circ H^a(x,t) = H^b(H^a(x,t),t)$ is a homotopy between $f^b \circ f^a$ and $g^b \circ g^a$.

1.3 Homotopies of Paths

When we are talking of paths we have a slightly stronger notion of homotopy- we fix the endpoints.

Definition 1.3. A *homotopy of paths* between paths $f, g : I \to X$ is a homotopy $H : I \times I \to X$ between f and g such that

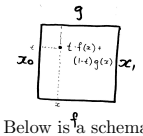
i. $f(0) = H(0,t) = g(0)$, and $f(1) = H(1,t) = g(1)$ for all $t \in I$.

Problem 1.5. Give an example (without proof) of two paths that are homotopic under Definition 1.2, but not under Definition 1.3

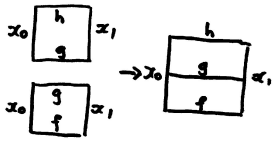
Hint:
Take Y to be the punctured plane $\mathbb{R} \setminus \{(0,0)\}$

From now on, when we say that two paths are *homotopic* we will always mean that there is a homotopy of paths between them. In this case, we write $f \simeq g$.

Example 1.4. Any two paths f and g in $\mathbb{R}$ with the same endpoints $f(0) = g(0) = x_0$ and $f(1) = g(1) = x_1$, are homotopic. Indeed the function $H : I \times I \to \mathbb{R}$ defined by$H(x,t) = tf(x) + (1-t)g(x)$ is a homotopy of paths between them.



In our figures above, we have drawn homotopies of paths as though $Y = \mathbb{R}^2$. Another way of drawing them is to simply draw $I \times I$ and label it with the value of the homotopy H. On the left, is a schematic of the homotopy H from example 1.4. Below is a schematic of the the product $H^a \cdot H^b$ of two homotopies of paths.



For a path f in a space X let $[f]$ be the class of f with respect to the equivalence relation $\simeq$, that is, let $[f]$ be the family of paths homotopic to f. The path operations of inverse and composition respect this equivalence.

Problem 1.6. Show that if $f \simeq g$ then $f^{-1} \simeq g^{-1}$. Show that if $g_1 \in [f_1]$ and $g_2 \in [f_2]$, then $g_1 \cdot g_2 \in [f_1 \cdot f_2]$.

Working with equivalence classes of paths instead of paths themselves, we are closer to having a group. For all $x_0 \in X$ let x_0 represent the constant path defined by $I \to X : x \to x_0$ for all x.

Problem 1.7. Show that $[f \cdot f^{-1}] = [f(0)]$ for any path f. Show that $[f(0) \cdot f] = [f]$. Is $[f \cdot f(0)] = [f]$? Is $[f^{-1} \cdot f] = [f(0)]$?

We still don't have an identity, or a well defined product, but for paths with the same endpoints we have a left identity. Lets take it one step further.

1.4 The Fundamental Group

To make a group out of the equivalence classes of paths, we restrict ourselves to paths that go to and from a fixed endpoint.

Definition 1.5. A path f is called a *loop* if $f(0) = f(1)$. The point $x_0 = f(0)$ is its *basepoint*.

For two loops f and g with the same basepoint, the product $[f][g] = [f \cdot g]$ is now well defined. So we will be able to make a group out of the set of loops with a given basepoint x_0.

Let

$$\pi_1(X, x_0) = \{[f] \mid f \text{ is a loop in } X \text{ with basepoint } x_0\}.$$

In this subsection, we prove the following.

Proposition 1.6. *Let x_0 be a point in the space X, then $\pi_1(X, x_0)$ is a group with product $[f][g] = [f \cdot g]$, identity $[x_0]$ and inverse $[f]^{-1} = [f^{-1}]$.*

This group is called the *fundamental group* of the space X at basepoint x_0. It is a topological invariant in the sense that if $h : X \to Y$ is a homeomorphism, then $\pi_1(X, x_0)$ is isomorphic to $\pi_1(Y, h(x_0))$.

Before we prove Proposition 1.6, we introduce a tool. Let $\text{inj} : X \to R$ be the (injection) path $\text{inj}(x) = x$. For any path $\phi : I \to I$ with $\phi(0) = 0$ and $\phi(1) = 1$, we can defining a homotopy of paths H_ϕ in I between inj and ϕ. So for any path f, $f \circ H_\phi$ is a homotopy of paths between $f = f \circ \text{inj}$ and $f \circ \phi$, and so $[f] = [f \circ \phi]$. The path $f \circ \phi$ is called a *reparameterisation of f*. We've seen reparametrisations in calculus.

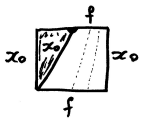
Now to prove Proposition 1.6 we have to show for all $[f], [g] \in \pi_1(X, x_0)$ that

- $[x_0][f] = [f] = [f][x_0]$ (identity),
- $[f]^1[f] = [x_0] = [f][f]^1$ (inverse), and
- $([f][g])[h] = [f]([g][h])$ (associativity).

To see that $[x_0][f] = [f]$, let $\phi : I \to I$ be the continuous (piecewise linear) function

$$\phi(x) = \begin{cases} 0 & \text{if } x \leq 1/2 \\ 2x - 1 & \text{if } x > 1/2 \end{cases}.$$

Then $x_0 \cdot f$ is the reparametrisation $f \circ \phi$ so is homotopic to f. The proof that $[f][x_0] = [f]$ is identical, so we have identity. The figure on the right suggests a different homotopy between f and $x_0 \cdot f$.



The proof of inverse follows from Problem 1.7. Complete the proof with the following problem.

Problem 1.8. Show that the product $[f][g] = [f \cdot g]$ is associative.

Hint:

Prop 1.6 is Prop 1.3 of [Ha]. This is done here

Problem 1.9. What is the group $\pi_1(R^2, (0, 0))$? (It is difficult to prove right now, but can you think of any space with a bigger fundamental group? ...with a non-abelian fundamental group?)

Problem 1.10. Show that the fundamental group is a topological invariant.

1.5 The fundamental group of the circle

In one of our exercises we saw that $\mathbb{R}^2$ has trivial fundamental group. Such spaces are called *simply connected*, and one could show that the real space $\mathbb{R}^i$ is

simply connected for all positive integers i. This name should suggest that the punctured plane, which is not simply connected by the other definitions we've seen, has non-trivial fundamental group. And it does. But how do we show this. It is difficult to show that two loops are not homotopic.

We look first at the fundamental group of the circle as it is the 'simplest' non-simply connected space. Indeed, $\pi_1(S^1)$ is isomorphic to $\mathbb{Z}$ via the mapping takes $[f]$ for a loop f to the number of times it wraps around S^1. (We call this its wrapping number) As believable as this is, its proof involves some complications. We have to show that the wrapping number is well defined, and preserved under homotopy.

First, let $p : \mathbb{R} \to S^1 : r \to e^{i2\pi r}$. A nice way to visualise p is to embed $\mathbb{R}$ into $\mathbb{C} \times \mathbb{R}$ as the parametrised curve $(e^{i2\pi r}, r)$ and then view p as the projection down to $\mathbb{C}$.

To show that the wrapping number of loop is well defined, we show that any loop 'lifts' uniquely via p^{-1} to a path in $\mathbb{R}$ whose endpoints differ by the wrapping number. Moreover, we show the wrapping number is preserved under homotopy by showing that a homotopy of paths lifts uniquely to a homotopy of paths in $\mathbb{R}$.

Lemma 1.7. *Let f be a path in S^1 with $f(0) = e^0$. Then there exists a unique path $\widetilde{f}$ in $\mathbb{R}$ with $\widetilde{f}(0) = e^0$ and $p \circ \overline{f} = f$.*

Proof. We will prove this in more generality when we talk of covering spaces, but for now, just observe that p is locally invertible, that is, for every x in S^1 and every $\widetilde{x}$ in $p^{-1}(x)$ there is an open set O around $\widetilde{x}$ such that p is a homeomorphism of O to $p(O)$. So once $\widetilde{f}(i)$ is uniquely determined, $\widetilde{f}(i + \varepsilon)$ is uniquely determined for any ε small enough that $(i + \varepsilon)$ is in $p(O)$. Thus $\widetilde{f}$ can be uniquely defined by induction, (you need to use the compactness of S^1 to show that $\widetilde{(f)}$ can be so defined in a finite number of steps). □

Slightly more generally we can show the following.

Lemma 1.8. *Let $H : I \times I \to S^1$ be a homotopy of paths with start point e^0. Then there exists a unique homotopy of paths $\widetilde{H}$ in $\mathbb{R}$ with $\widetilde{H}(I, 0) = e^0$ and $p \circ \widetilde{H} = H$.*

Proof. Basically, apply Lemma 1.7 to each path $H(0, \cdot)$, and show the lift $\widetilde{H}$ that you get in this way is continuous. □

Theorem 1.9. $\pi_1(S^1) \cong \mathbb{Z}$.

Proof. Define $\phi : \pi_1(S^1) \to \mathbb{Z}$ by $\phi([f]) = \widetilde{f}(1)$ for a loop f in S^1.

This is a map to $\mathbb{Z}$ because $f(1) = e^0$, and so $p^{-1}(f(1)) = in\mathbb{Z}$. To see that it is well defined, observe that for $f \simeq g$, the homotopy H between f and g lifts, by Lemma 1.8, to a homotopy of loops between $\widetilde{f}$ and $\widetilde{g}$. So $\widetilde{f}(1) = \widetilde{g}(1)$.

The fact that $\phi([\omega_n]) = n$ where ω_n is the loop $t \to e^{i2\pi nt}$ in S^1, gives us the surjectivity of ϕ. Injectivity follows from the fact that for any loop f with $\phi([f]) = n$, $\widetilde{H}(x,t) = (1-t)\widetilde{f}(x) + t(\widetilde{\omega_n}(x))$ is a homotopy between $\widetilde{f}$ and $\widetilde{\omega_n}$ and so (as we will see at the start of the next section) $p \circ \widetilde{H}$ is a homotopy in S^1 between f and ω_n. Clearly $\omega_m \cdot \omega_n = \omega_{m+n}$, and so (recall that $\mathbb{Z}$ is an additive group)

$$\begin{aligned} \phi([\omega_m] + [\omega_m]) &= \phi([\omega_m \cdot \omega_n]) \\ &= \phi([\omega_{m+n}]) \\ &= m + n \\ &= \phi([\omega_m]) + \phi([\omega_n]). \end{aligned}$$

Thus ϕ is a homomorphism, and so an isomorphism. □

2 Homotopy Equivalence

2.1 Induced Homomorphism of Fundamental Groups

Given a continuous function $\phi : X \to Y$ (of path connected spaces), and a loop f in X with basepoint x_0 the composition $\phi \circ f$ is a loop in Y with basepoint $\phi(x_0)$.

Now ϕ is well defined on fundamental groups: if H is a homotopy between loops f and g with basepoint x_0 then $H \circ f$ is a homotopy between loops $\phi \circ f$ and $\phi \circ g$. So ϕ *descends to homotopy*, that is it induces a well defined map

$$\phi_* : \pi_1(X, x_0) \to \pi_1(Y, \phi(x_0)) \mapsto [\phi \circ f]$$

between the fundamental groups.

The following properties are basic.

Proposition 2.1.

i. For continuous function $\phi : X \to Y$, ϕ_* is a homomorphism of groups.

ii. For $\phi : X \to Y$ and $\psi : Y \to Z$, $(\psi \circ \phi)_* = \psi_* \circ \phi_*$.

iii. Where $\mathrm{id} : X \to X$ is the identity on X, $\mathrm{id}_* : \pi_1(X, x_0) \to \pi_1(X, x_0)$ is also identity.

iv. If $\phi : X \to Y$ is a homeomorphism, then ϕ_* is an isomorphism.

Proof. Part (i) is directly from the definition of the product of paths. Indeed, it gives

$$\phi \circ (f \cdot g)(i) = \left\{ \begin{array}{ll} \phi \circ f(2i) & \text{if } i \leq 1/2 \\ \phi \circ g(2i-1) & \text{if } i > 1/2 \end{array} \right\} = (\phi \circ f) \cdot (\phi \circ g)(i),$$

and so we have that

$$\begin{aligned} \phi_*([f][g]) &= \phi_*([f \cdot g]) = [\phi \circ (f \cdot g)] = [(\phi \circ f) \cdot (\phi \circ g)] \\ &= [(\phi \circ f)][(\phi \circ g)] = \phi_*([f])\phi_*([g]) \end{aligned}$$

Parts (ii) and (iii) are just as immediate from definitions. From them we show (iv). Indeed, if ϕ is a homeomorphism, then it is continuous, and has inverse ϕ^{-1} which is also continuous. By part (i) ϕ_* and ϕ_*^{-1} are homomorphisms and by (ii) and (iii) we have that

$$\phi_* \circ \phi_*^{-1} = (\phi \circ \phi^{-1})_* = \mathrm{id}_*,$$

and similiarily that $\phi_*^{-1} \circ \phi_* = \mathrm{id}_*$, are the identity. So ϕ_* is an isomorphism. □

Note, items (i - iii) of this proposition are the statement that π is a functor between the categories of bases spaces and groups. Item (iv) then follows as a property of functors. We will explain this later when we talk of category theory.

Problem 2.1. If X is a subspace of Y, what can you say about the relation of $\pi_1(X, x_0)$ and $\pi_1(Y, x_0)$ for x_0 in X?

Proposition 2.2. *If $x_0 \in X$ and $y_0 \in Y$, then $\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$.*

Proof. Recall that a map $f : S \to X \times Y$ is continuous if and only if its composition with the projections $p_1 : X \times Y \to X : (x,y) \to x$ and $p_2 : X \times Y \to X : (x,y) \to y$ are continuous. So a loop f in $X \times Y$ with basepoint (x_0, y_0) induces loops $p_1(f)$ in X and $p_2(f)$ in Y. Similarily a homotopy H from $f' \in [f]$ to f induces homotopies $p_1(H)$ from $p_1(f')$ to $p_1(f)$ and $p_2(H)$ from $p_2(f')$ to $p_2(f)$. So the map

$$\pi_1(X \times Y, (x_0, y_0)) \to \pi_1(X, x_0) \times \pi_1(Y, y_0) : [f] \mapsto ([p_1(f)], [p_2(f)])$$

is well defined. One easily checks that it is a homomorphism. To see that it is surjective observe that for loops f in X with basepoint x_0 and g in Y with basepoint y_0 the (pullback) map $f \times g : I \to X \times Y : t \mapsto (f(t), g(t))$ is a loop in $X \times Y$ with basepoint (x_0, y_0). To see that the group map in injective, we can pull back homotopies in X and Y in the same way. □

Example 2.3. The torus $T = S^1 \times S^1$ has fundamental group $\mathbb{Z} \times \mathbb{Z}$.

2.2 Homotopy Equivalence

We have seen that if the continuous function ϕ between spaces is a homeomorphism, then the corresponding homomorphism ϕ_* of fundamental groups is an isomorphism. There is however a weaker condition on ϕ that implies that ϕ_* is an isomorphism.

The statement that continuous $\phi : X \to Y$ is a homeomorphism is the statement that there exists $\psi : Y \to X$ such that $\phi \circ \psi$ and $\psi \circ \phi$ are the identity maps. Weakening this, we have:

Definition 2.4. A continuous function $\phi : X \to Y$ is a *homotopy equivalence* if there exists a continuous function $\psi : Y \to X$ such that both $\psi \circ \phi$ and $\phi \circ \psi$ are homotopic to the identity. If there is a homotopy equivalence between two spaces, they are called *homotopy equivalent.*

The following is a key idea.

Proposition 2.5. *If $\phi : X \to Y$ is a homotopy equivalence, then for any x_0 in X, $\pi_1(X, x_0) \cong \pi_1(Y, \phi(x_0))$.*

Proof of a special case. Right now we prove the proposition with one stronger assumption: that the homotopy in X between id_X and $\psi\phi$ fixes x_0, and the homotopy in Y between id_Y and $\phi\psi$ fixes $\phi(x_0)$.

For any continuous $\phi : X \to Y$ and $\psi : Y \to X$, the sequence

$$X \xrightarrow{\phi} Y \xrightarrow{\psi} X \xrightarrow{\phi} Y$$

induces the sequence

$$\pi_1(X, x_0) \xrightarrow{\phi_*} \pi_1(Y, \phi(x_0)) \xrightarrow{\psi_*} \pi_1(X, \psi\phi(x_0)) \xrightarrow{\phi_*} \pi_1(Y, \phi\psi\phi(x_0))$$

If there is a homomopy of id_X to $\psi\phi$ to which fixes x_0 then by composition, ones gets a homotopy of paths between f and $\psi\phi(f)$ for any loop f in X based at x_0. This gives that

$$\psi_* \circ \phi_*([f]) = [\psi \circ \phi(f)] = [f],$$

and so $\psi_* \circ \phi_*$ is the identity. Similarily if there is a homotopy of id_Y to $\phi\psi$ that fixes $\phi(x_0)$, then $\phi_* \circ \psi_*$ is the identity. It follows that ϕ_* is an isomorphism. $\square$

However, in general, the homotopies implied by the fact that ϕ is a homotopy equivalence need not fix any basepoint x_0. So to prove Proposition 2.5, we have to do some work. We complete this proof in the next section.

Definition 2.6. Let $id : Y \to Y$ be the identity map, and for subspace X of Y, let $r : Y \to X$ be a *retraction* of Y to X, that is, a continuous map that fixes every x in X. If there is a homotopy between id and r in Y which fixes X for all t, (such a homotopy is called a *homotopy relative to* X) then we say Y *deformation retracts* to X. A space is *contractible* if it deformation retracts to a point.

Problem 2.2. Show that a deformation retraction is a homotopy equivalence.

Problem 2.3. Show that the punctured plane $\mathbb{R}^2 \setminus (0,0)$ deformation retracts to S^1, so where x is any point in $\mathbb{R}^2$, $\pi_1(\mathbb{R}^2 \setminus \{x\}) \cong \mathbb{Z}$.

Corollary 2.7. *If Y deformation retracts to X, then for every x_0 in X, $\pi_1(X, x_0) \cong \pi_1(Y, \phi(x_0))$. In particular, if Y is contractible, then $\pi_1(Y, y_0) = 0$ for any y_0 in Y.*

2.3 Independence of Basepoints

A space X is *path connected* if for any two points x_0 and y_0, there is a path b from x_0 to y_0. Given a loop f with basepoint x_0, $b^{-1} \cdot f \cdot b$ is a loop with basepoint y_0. This induces a mapping $\beta_b : \pi_1(X, x_0) \to \pi_1(X, y_0)$ by $\beta_b([f]) = [b^{-1} \cdot f \cdot b]$. To see that this mapping is well defined on the fundamental groups, we observe that if $\{H_t\}$ is a homotopy between f and g, then $\{b^{-1}H_t b\}$ is a homotopy between $b^{-1}fb$ and $b^{-1}gb$. To see that it is a homomorphsims of groups, observe that $\beta_b([f][g]) = \beta_b([f \cdot g]) = [b^{-1}fgb] = [b^{-1}fbb^{-1}gb] = [b^{-1}fb][b^{-1}gb] = \beta_b([f])\beta_b([g])$. To see that it is an isomorphism, observe that $\beta_{b^{-1}}$ is its inverse.

This isomorphism commutes with homotopies.

Lemma 2.8 (Lemma 1.19 of [Ha]). *Let $H : X \times I \to Y$ be a homotopy between $f = H(\cdot, 0)$ and $g = H(\cdot, 1)$. For $x_0 \in X$ let $b = H(x_0, \cdot)$ be the induced path between $f(x_0)$ and $g(x_0)$. Then $g_* = \beta_b \circ f_*$. That is, the diagram below commutes.*

$$\begin{tikzcd}
 & \pi_1(X, x_0) \arrow[dl, "f_*"'] \arrow[dr, "g_*"] & \\
\pi_1(Y, f(x_0)) \arrow[rr, "\beta_b"] & & \pi_1(Y, g(x_0))
\end{tikzcd}$$

Proof. For each $t \in I$ let b_t be the path $b_t(i) = b(t + (1-t)i)$ from $b(t)$ to $b(1)$ along b. For a loop p in X, with basepoint x_0, we have that $\{b_t^{-1}(H_t \circ p)b_t\}_I$ is a homotopy of paths between the loops $b^{-1}(f \circ p)b$ and $(g \circ p)$ in Y. So $g_*([p]) = [g \circ p] = [b^{-1}(f \circ p)b] = \beta_b \circ f_*([p])$, as needed. $\square$

This allows us to finish the proof of Proposition 2.5. Indeed, letting b be the path in X between x_0 and $\psi\phi(x_0)$ which is induced by the homotopy between id_X and $\psi\phi : X \to X$, the lemma gives us that the following commutes.

$$\begin{tikzcd}
 & \pi_1(X, x_0) \arrow[dl, "\mathrm{id}_*"'] \arrow[dr, "\psi_* \circ \phi_*"] & \\
\pi_1(X, x_0) \arrow[rr, "\beta_b"] & & \pi_1(X, \psi\phi(x_0))
\end{tikzcd}$$

As id_* and b_* are isomorphisms, so is $\psi_*\phi_*$, so ϕ_* is injective. Similarly we can show that $\phi_*\psi_*$ is an isomorphism, so ϕ_* is surjective. This finishes the proof.

We have shown that in a path connected space, the fundamental group is independent of basepoint. We will generally restrict ourselves to looking at path connected spaces, and will therefore write $\pi_1(X)$ for the fundamental group $\pi_1(X, x_0)$, where x_0 is any point in X.

2.4 Other Homotopy Groups

The group $\pi_1(X)$, for a path connected space X, catalogues the homotopy classes of embeddings of the one-sphere S_1 into X. This is useful in seperating say the torus from the plane, but is no help in differentiating the plane from the sphere (as we will see.)

In the same way, we can define the group $\pi_i(X)$ for any integer $i \geq 1$ to catalogue the homotopy classes of the embeddings of S_i into X. These invariants will separate many spaces that have the same fundamental group, though cannot separate homotopy equivalent spaces.

The higher homotopy groups are hard to calculate. Instead of spending time with them we will work with a weaker, but more easily compute invariant called homology. Before we do this, we look at some of the main theorems of homotopy theory. These will have analogues in homology theory.

3 Fun with homotopy Theory

We will investigate in more detail some of the ideas we saw in the proof of Theorem 1.9. But before we do, we look at a couple consequences of this theorem.

3.1 The Fundamental Theorem of Algebra

Theorem 3.1 (The Fundamental Theorem of Algebra)**.** *Every non-constant polynomial over $\mathbb{C}$ has a root in $\mathbb{C}$.*

Proof. Let $p(z) = z^n + a_{n-1}z^{n-1} + \dots a_1 z + a_0$ be a polynomial in $\mathbb{C}$ with no roots. Then

$$f_q(s) = \frac{p(qe^{i2\pi s})/p(q)}{|p(qe^{i2\pi s})/p(q)|}$$

is a loop in S^1 based at 1 for all q. Letting q vary from 0 to any r we get a homotopy of loops between constant 1 and f_r. So $[f_r]$ is zero in $\pi_1(S^1)$.

Now take r bigger than $|a_0| + |a_1| + \dots + |a_{n-1}|$ we have that for z in the circle $re^{i2\pi t}$ $(0 \leq t \leq 1)$,

$$\begin{aligned} |z^n| = r^n = r \cdot r^{n-1} &> (a_0 + a_1 + \dots a_{n-1}) \cdot |z^{n-1}| \\ &> |a_{n-1}z^{n-1} + \dots a_1 z + a_0|. \end{aligned}$$

So $p_t(z) = z^n + t(a_{n-1}z^{n-1} + \dots a_1 z + a_0)$ has no zeros on the circle $re^{i2\pi t}$. Thus letting t go from 1 to 0

$$f_t(s) = \frac{p_t(re^{i2\pi s})/p_t(r)}{|p_t(re^{i2\pi s})/p_t(r)|}$$

is a homotopy of loops from f_r to ω_n. As $[f_r]$ is zero, and $[\omega_n] = n$, we have that the degree n of p is 0, so p is constant. □

3.2 The Brouwer fixed point Theorem

Where D^2 is the unit disk $\{x, y \mid x^2 + y^2 \leq 1\}$, one can show as we did for $\mathbb{R}$, that $\pi_1(D^2)$ is trivial. Let $i : S^1 \to D^2$ be the inclusion map. There can be no continuous map r such that $r \circ i$ is the identity on S^1, (that is, such that r is the identity on S^1,) or else it would induce an isomorphism $r_* \circ i_* : \pi_1(S^1) \to \pi_1(D^2) \to \pi_1(S^1)$ of $\pi_1(S^1)$ that factors through the trivial group $\pi_1(D^2)$, which is impossible. With this, we get the following.

Theorem 3.2 (The Brouwer Fixed Point Theorem)**.** *Any continuous map $f : D^2 \to D^2$ has a fixed point.*

Proof. Assume there is a continuous map f with no fixed point. Define $r : D^2 \to S^1$ as follows. For $x \in D^2$, we have that $f(x) \neq x$. Take the ray in D^2 starting at $f(x)$ and passing through x. Let $r(x)$ be the point at which it intersects S^1. One can show that r is continuous, and clearly for $s \in S^1$, $r(s) = s$, so r is the identity on S^1. We saw above that this is impossible. □

Problem 3.1. Show that r is continuous.

Hint:

For $z = (x, y)$ in D^2 let $f(z) = (f_x(z), f_y(z))$, and $r(z) = (r_x(z), r_y(z))$. Define $r_x(z)$ in terms of $x, y, f_x(z)$, and $f_y(z)$. As these are all continuous, so is $r_x(z)$.

3.3 Covering Spaces

In this section, we make the assumption that all spaces are locally path connected (excluding such things as the topologists comb,) and *semi-locally simply connected*, which means that for every point, there is an open neighbourhood around the point with trivial fundamental group.

Definition 3.3. A continuous function $p : \widetilde{X} \to X$ is a *covering* of a space X if it satisfies the following. There is an open cover C of X such that for each $U \in C$, p^{-1} is the union of disjoint open sets in $\widetilde{X}$ each of which p maps homeomorphically to U. The space $\widetilde{X}$ is called a covering space of X.

The function $p : \mathbb{R} \to S^1 : r \to e^{i2\pi r}$ from the last section is a covering of S^1. Indeed fix small $\varepsilon > 0$ and consider the open sets $U_1 = \{e^{i\theta} \mid \theta \in (-(pi/2 + \varepsilon), \pi/2 + \varepsilon)\}$ and $U_2 = \{e^{i\theta} \mid \theta \in ((pi/2 - \varepsilon), 3\pi/2 + \varepsilon)\}$. Then $C = \{U_1, U_2\}$ is an open cover of S^1 satisfying the condition of the definition.

There are other covering spaces of S^1 too. S^1 with $p : e^{ir} \mapsto e^{inr}$ is one. And the disjoint union of two covering spaces is a covering space. So we will usually only care about simply connected covering spaces. Several copies of S^1 with p being the identity on each is one.

Definition 3.4. For a function $f : Y \to X$, a lift $\widetilde{f}$ of f to a covering $p : \widetilde{X} \to X$ is a function $\widetilde{f} : Y \to \widetilde{X}$ such that $f = p \circ \widetilde{f}$.

Lemma 3.5. *Let $\widetilde{X}$, p be a covering space of X. Then for any path f in X with $f(0) = x_0$, and any lift $\widetilde{x_0} \in p^{-1}(x_0)$, there exists a unique lift $\widetilde{f}$ of f in $\widetilde{X}$ with $\widetilde{f}(0) = \widetilde{x_0}$.*

Proof. Let f be a path in X with $f(0) = x_0$ and $\widetilde{x_0} \in p^{-1}(x_0)$. As f is compact, one can find $0 = t_0 < t_1 < t_2 < \cdots < t_n = 1$ such that there for each i there is an open set U_i in the open cover C of Defintion 3.3 such that $\{f(t) \mid t \in [t_i, t_{i+1}]\} \subset U_i$.

Let $\widetilde{f}(0) = \widetilde{x_0}$. Assume that there is a unique lift $\widetilde{f}([t_0, t_i])$ of $f([t_0, t_i])$ in $\widetilde{X}$ with $\widetilde{f}(0) = \widetilde{x_0}$. Then there is unique open $\widetilde{U_i}$ in $\widetilde{X}$ with $\widetilde{f}(t_i) \in \widetilde{U_i}$ and having $p(\widetilde{U_i}) = U_i$. So letting $\widetilde{f}(t) = p^{-1}(f(t)) \cap \widetilde{U_i}$ for each $t \in [t_i, t_{i+1}]$ is the unique extension of $\widetilde{f}([t_0, t_i])$ to a lift of $f([t_0, t_{i+1}])$ in $\widetilde{X}$. The lemma follows by induction. □

With a similar proof, one can extend this to the following homotopy lifting property, which implies that homotopic paths $f \simeq g$ lift to homotopic paths $\widetilde{f} \simeq \widetilde{g}$ in a covering space.

Lemma 3.6 (Homotopy Lifting Property)**.** *Let $p : \widetilde{X} \to X$ be a covering. For any homotopy of paths H with $H(i, 0) = x_0$ (for all i), and any $\widetilde{x_0} \in p^{-1}(x_0)$ there is a unique lift homotopy of paths $\widetilde{H}$ in $\widetilde{X}$ with $\widetilde{H}(i, 0) = \widetilde{x_0}$.*

Now for a covering $p : \widetilde{X} \to X$, we have, of course

$$p_* : \pi_1(\widetilde{X}, \widetilde{x_0}) \to \pi_1(X, x_0).$$

If $p_*([f]) = 0$ for some loop in $\widetilde{X}$, then the nullhomotopy of $p \circ f$ lifts to a nullhomotopy of f in $\widetilde{X}$ by the homotopy lifting property, and so p_* is injective. Non-nullhomotopic loops in X lift to non-nullhomotopic loops, or non-loops (between $\widetilde{x_0}$ and other elements of $p^{-1}(x_0)$), in $\widetilde{X}$. The elements of $p^*(\pi_1(\widetilde{X}) \subset \pi_1(X)$ are the ones that lift to non-nullhomotopic loops.

With this observation, we can prove the following

Lemma 3.7 (Lifting Criterion)**.** *Let $p : \widetilde{X} \to X$ be a cover with $p(\widetilde{x_0}) = x_0$, and $f : Y \to X$ be a continuous function with $f(y_0) = x_0$. Then f lifts to $\widetilde{f}$ in $\widetilde{X}$ if and only if $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\widetilde{X}, \widetilde{x_0}))$.*

Proof. 'Only if' is clear because if $\widetilde{f}$ exists, then

$$f_*(\pi_1(Y, y_0)) = p_* \circ \widetilde{f}_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\widetilde{X}, \widetilde{x_0})).$$

So assume that $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\widetilde{X}, \widetilde{x_0}))$. We define $\widetilde{f}$. We may assume that $\widetilde{f}(y_0) = \widetilde{x_0}$. For any other $y \in Y$, we take a path γ in Y from y_0 to y, and let $\widetilde{y}$ be the end of the unique lift of $f(\gamma)$ (with $f(\gamma(0)) = x_0$) to $\widetilde{X}$ with initial point $\widetilde{x_0}$. If this is not well defined then there are different paths γ and γ' between y_0 and y, so a loop $l = \gamma^{-1}\gamma'$ in Y, such that the loop $f(l)$ in X lifts to a non-path in $\widetilde{X}$ with endpoint $\widetilde{x_0}$. But we observed before that $[f(l)] \in f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\widetilde{X}, \widetilde{x_0}))$ lift to loops (non-nullhomotopic ones) in $\widetilde{X}$. This is a contradiction. □

Another fact: $|p^{-1}(x)|$ is constant (as X is connected) and one can show that it is the index $|\pi_1(X, x_0)|/|\pi_1(\widetilde{X}, \widetilde{x_0})|$ of $p^*(\pi_1(\widetilde{X}, \widetilde{x_0})$ in $\pi_1(X, x_0)$. This number is called the number of *sheets* of the cover.

Simplifying statements by assuming that everything is path connected, we have that for an n sheet cover $p : \widetilde{X} \to X$ and an embedding $f : Y \to X$, f lifts to commuting $\widetilde{f}$ if and only if $f_*(\pi_1(Y)) \subset p_*(\pi_1(\widetilde{X})$.

$$\begin{tikzcd} Y \arrow[r, "\widetilde{f}"] \arrow[dr, "f"'] & \widetilde{X} \arrow[d, "p"] \\ & X \end{tikzcd}$$

In particular, every covering space of X corresponds to a subgroup of $\pi_1(X)$ via π_1. One can show:

- covers with the same or conjugate corresponding subgroup are isomorphic, (conjugate subgroups correspond to different choices of base point)
- for every subgroup there is a corresponding cover.

The cover corresponding to the trivial subgroup is necessarily simply connected (it fundamental group is trivial). It is called the universal cover, because by

the lifting criterion, it has the universal property of factoring through any other cover.

Viewing the covers as a heirarchy with X itself on the bottom, and the universal cover on the top, we can view going up the heirarchy as choosing non-nullhomotopic loops, and breaking them open.

3.4 Borsuk-Ulam Theorem

Theorem 3.8. *Every continuous function $f : S^2 \to R^2$ has a pair of antipodal points, that is, there is $x \in S^2$ such that $f(-x) = f(x)$.*

Proof. Assume that there is a function $f : S^2 \to R^2$ without antipodal points. Then the function

$$g : S^2 \to S^1 : x \mapsto \frac{f(x) - f(-x)}{|f(x) - f(-x)|}$$

is continuous and satisfies $g(x) = -g(-x)$ for all x. In particular, its restriction to any equator is a loop in S^1 satisfying $g(x) = -g(-x)$.

By continuity and the property $g(x) = -g(-x)$, (which translates to $l(x) = -l(x + 1/2)$ for $x \in [0, 1/2]$ when viewed as a map l from I), one can show that it lifts to a non-loop in the covering space $p : \mathbb{R} \to S^1 : r \to e^{i2\pi r}$ of S^1, so is non-nullhomotopic. But this contradicts the fact that it is the image of a loop in S^2, in which all loops are null-homotopic. (See page 33 of [Ha] for more details.) □

One could prove the same for higher dimensions, that every continuous function $f : S^n \to R^n$ has a pair of anti-podal points.

3.5 Van Kampens Theorem

Van Kampens Theorem allows us to compute the fundamental group of space in terms of certain subspaces.

As a motivating example, consider the space $A \bigvee B$ consisting of two copys A and B of S^1 with intersection the point x_0. This is called the *wedge sum* of A and B at the common point x_0. The loops that go around A once or B once, in either direction, are non-nullhomotopic. Call them respectively a, a^{-1}, b, and b^{-1}. The loops aa, aaa, ab, aba^{-1} are distinct and non-nullhomotopic, but aa^{-1} is nullhomotopic. Similarily $abb^{-1}a$ is homotopic to aa. The fundamental group of $A \bigvee B$ is the *free product* of the fundamental groups of A and B.

Problem 3.2. Show the loop a in $A \bigvee B$ is non-nullhomotopic.

Given two groups G and G', the *free product* $G \star G'$ is the group defined as follows. A *word* over G and G' is a string $g_1 g_2 \dots g_n$ of finite length in which each g_i belongs to G or G'. A word is *reduced* by removing instances of the identity elements of G or G', and replacing $g_i g_{i+1}$ with its product if both are in the same group G or G'. The elements of $G \star G'$ are the reduced words, the identity is the empty word, and the operation is concatenation followed by reduction.

Clearly both G and G' are subgroups of $G \star G'$ via the injections $\mathrm{inj}_G : G \to G \star G' : g \to g$ and $\mathrm{inj}_{G'} : G' \to G \star G' : g \to g$ and for homorphisms $\phi : G \to H$ and $\phi' : G' \to H$ it is a property of the star product that there is a unique homomorphism of $G \star G' \to H$ that commutes with ϕ and ϕ'. It takes a reduced word $g_1 g_2 \dots g_n$ to $\phi^*(g_1)\phi^*(g_2) \dots \phi^*(g_n)$ where ϕ^* is ϕ or ϕ' depending on whether g_i is from G or G'.

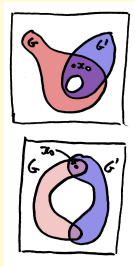
If a space X is the union of two spaces A and B, and x_0 is in their intersection, then any loop based at x_0 in A or B is in X, so we uniquely get

$$\Phi : \pi_1(A, x_0) \star \pi_1(B, x_0) \to \pi_1(X, x_0).$$

commuting with duals of the inclusion maps.

Problem 3.3. Show that Φ is not in general injective or surjective.

Hint:



As we see in the above problem, loops in the intersection can of A and B contribute to the kernel of Φ: the same loop gets the name f_A in A and f_B in B, but clearly $f_A \cdot f_B^{-1}$ is nullhomotopic.

In terms of Φ this means that for $[f] \in \pi_1(A \cap B)$ where $[f_a] \in \pi_1(A)$ and $[f_b] \in \pi_1(B)$ are induced by the injection of $A \cap B$ into A and B, $[f_a][f_b]^{-1}$ is in the kernel of Φ.

Van Kampen's Theorem gives conditions under which Φ is surjective, and under which the kernel is exactly those elements we've discussed. Further, it does so in the more general situation when X is composed of more than two spaces.

Theorem 3.9. *Let X be the union of a set of path connected open sets A_{iI} each containing x_0. If $A_i \cap A_j$ is path connected for each pair i, j, then the natural map*

$$\Phi : \star_i \pi_i(A_i, x_0) \to \pi_1(X, x_0)$$

induced by their inclusion maps into X, is surjective. Furthermore, if $A_i \cap A_j \cap A_k$ is path connected for every triple i, j, k, then the kernel of Φ consists of the subgroup generated by elements of the form $[f_i][f_j]^{-1}$ where $[f] \in \pi_1(A_i \cap A_j, x_0)$ and f_i and f_j are the images of f under the inclusions of $A_i \cap A_J$ in A_i and A_j respectively.

Problem 3.4. Prove the surjectivity in the case that $X = A_1 \cup A_2$.

Hint:

[Ha] page 44

PART II: HOMOLOGY

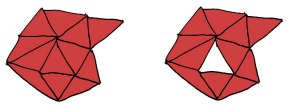
Homotopy is nice but is difficult to calculate. Homology is a similar invariant that is not as good in the sense that it will not distintinguish all pairs of spaces that homotopy will, but is easier to calculate. We start by defining a simple version of it on a relatively simple class of spaces called simplicial complexes. We then then look at how the algebra involved can be generalised, and then apply this algebra to more robust versions of homology.

4 Simplicial Homology

Consider discretising the theory of the fundamental group in the following way. Take a surface, and partition it into triangles. Instead of all the loops in the surface, consider only going along the edges of the triangles. We can deform a loop by adding the loop around a triangle which shares an edge in the other direction:

The homology class of a loop is all other loops we can get from such deformations. The first homology group is made up of these classes. The first striking simplificaton we see over the theory of homotopy, is that once our triangulisation is set, there is no mention of continuity. Things are beginning to feel more like algebra than topology.

Now I say 'triangles' above, but there is more to it than just that our theory should differentiate the following:



We will think in many dimensions, and worry about the interaction between dimensions.

4.1 Simplicial Complexes

We want to deal with more than just surfaces, so we need a higher dimensional analogue of triangles.

A *linear d-simplex* $\Delta^d = [v_0, \ldots, v_d]$ is the set

$$\{\sum_{i=0}^{d} a_i v_i \mid \sum_i a_i = 1; \forall i,\ a_i > 0\}$$

of points in the minimum convex solid in $\mathbb{R}^{d+1}$ containing $d+1$ points $v_0, \ldots, v_d$ in *general position* (i.e., linearly independent as vectors in $\mathbb{R}^{d+1}$). A d-simplex is a subspace of $\mathbb{R}^{d+1}$. (We use square brackets in $[v_0, \ldots, v_d]$, as the order of the points will be important.)

Removing one vertex v_i from a d-simplex $[v_0, v_1, \ldots, v_d]$ we get a $(d-1)$-simplex

$$[v_0, v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_d],$$

called a *face* of the d-simplex. The union of the faces of a d-simplex Δ^d is called the *boundary* of Δ^d and is denoted $\delta\Delta^d$. As it saves a couple of keystrokes, we might use the following notation:

$$[v_0, v_1, \ldots, \widehat{v_i}, \ldots, v_d] = [v_0, v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_d].$$

Thus a filled triangle is a 2-simplex $[v_0, v_1, v_2]$. Its boundary consists of the three 1-simplices $[v_1, v_2]$, $[v_2, v_3]$ and $[v_3, v_1]$. The boundaries of these edges, or 1-simplices, are the vertices, or 0-simplices $[v_1]$, $[v_2]$ and $[v_2]$.

The interior $\operatorname{Int}\Delta$ of a simplex Δ is $\Delta - \delta\Delta$.

As we triangualised a surface above, we apply a simplicial structure on a space by mapping a family of simplices onto it.

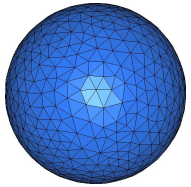


Image from the CGAL user manual

http://www.cgal.org/Manual/3.4/doc_html/cgal_manual/contents.html

For a space X, and each integer $d \geq 0$, let $\Delta^d(X)$ be a collection of continuous maps from d-simplices to X. The collection $\Delta(X) = \bigcup_d \Delta^d(X)$ of all of these maps is a Δ-*complex structure on* X if it satisfies the following.

- For each d, and each $\sigma \in \Delta^d(X)$, the restriction $\operatorname{Int}\sigma$ of σ to $\operatorname{Int}\Delta^d$ is injective. (Note that $\operatorname{Int}\Delta^0 = \Delta^0$.)
- Each vertex in X is in the image of $\operatorname{Int}(\sigma)$ for exactly one $\sigma \in \Delta(X)$.
- For each integer $d \geq 1$, each $\sigma \in \Delta^d(X)$, and each face F of Δ^d, the restiction $F|_\sigma$ of σ to F is in $\Delta^{d-1}(X)$.
- A set $A \subset X$ is open if and only if $\sigma^{-1}(A)$ is open for each $\sigma \in \Delta(X)$.

Where $\sigma \in \Delta^d(X)$ and F is a face of Δ^d, $\sigma|_F$ is a *face* of σ. The *boundary* $\delta\sigma$ of σ is the union of its faces.

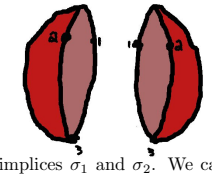
So far all the d-simplices we have talked about have been linear d-simplices. It is convenient to refer to a continuous map σ from a linear d-simplex to a space X as a (non-linear) d-simplex (in X). We will also identify σ with its image, and call its image a d-simplex.

This gives the definition of a Δ-complex structure $\Delta(X)$ a more intuitive description. We view it as building X up out of simplices as follows. Start with a bunch $\Delta^0(X)$ of 0-simplices in X. This is called the 0-skeleton of the structure. $\Delta^1(X)$ is a set of 1-simplices in X with their boundary 0-simplices identified with the 0-simplices in $\Delta^*(X)$. (Identify means take the appropriate quotient space of their union.) This is the 1-skeleton. Continue in this matter for increasingly higher sets in $\Delta(X)$.

For example, one can construct a Δ-complex structure on a cone from two 0-simplices, two 1-simplices, and a 2-simplex as follows.

The important information about a Δ-complex structure is the relationship between the different simplices, and so can be described algebraically in terms of the 0-simplices.

Letting the two 0-simplices in the above Δ-simplex structure of the cone be $[v_1]$ and $[v_2]$, we can describe the 1-simplices as $[v_1, v_2]$ and $[v_1, v_1]$. The 2-simplex is $[v_1, v_1, v_2]$. The boundary $\delta[v_1, v_1, v_2]$ of the 2-simplex consists of $[v_1, v_1]$ and two copies of $[v_1, v_2]$.



Consider the following Δ-complex structure on a sphere. Take two 2-complexes and identify their boundaries as defined by the labelling of their vertices in the figure on the left. This triangulisation of the sphere uses three 0-simplices: $[1], [2]$, and $[3]$, three 1-simplices $[1, 2]$, $[2, 3]$ and $[1, 3]$, and two 2-simplices σ_1 and σ_2. We cannot uniquely describe them by their vertices but we will want to refer to their vertices. We denote them $\sigma_1 = [1, 2, 3]_{\sigma_1}$ and $\sigma_2 = [1, 2, 3]_{\sigma_2}$. In general we denote the face of $[v_0, \ldots, v_d]_\sigma$ which omits the vertex v_i by $[v_0, \ldots, \widehat{v_i}, \ldots v_d]_\sigma$. So as in this example, a simplex such as $[1, 2]$ could be refered to as both $[1, 2]_{\sigma_1}$ and $[1, 2]_{\sigma_2}$.

A space has any number of different Δ-complex structures. One in which no two simplices in $\Delta(X)$ have the same set of vertices is called a simplicial complex structure. These are notationally simple, but tend to require more simplices so become harder to work with. A simplicial complex structure on S^1 needs at least four 2-simplices, compared to two for a Δ-complex structure. But we'll put up with this for now.

Problem 4.1. Find a simplicial complex structure for the Torus. What is the fewest number of simplices you can do it with? How about for a Δ-complex structure?

4.2 Simplicial Homology

Given a Δ-complex structure $\Delta(X)$ of a space X we construct a series $H_*^\Delta(\Delta(X))$ of groups $H_i^\Delta(\Delta(X))$, for $i \geq 0$, called the *simplicial homology* groups.

Remark 4.1. The superscript Δ distinguishes them from *singular homology* groups that we will talk about later. The two largely correspond, so we will only use the superscript when we want to distinguish them. In particular, we largely drop it in the next two sections, before we introduce singular homology. Further, it will turn out that the groups are independent of the choice of Δ-complex structure, and so we write them as $H_i(X)$.

We define these groups now for simplicial complex structures.

Let $\Delta(X)$ be a Δ-complex structure for a space X. For each $d \geq 0$ let $C_d = C_d(X)$ be the free abelian group generated by elements of $\Delta^d(X)$. Elements of C_d are (finite) linear combinations of d-simplices $\sum_\alpha a_\alpha \sigma_\alpha$ with coefficients a_α in $\mathbb{Z}$. Elements of C_d are called *d-chains*.

We define the boundary map $\delta_d : C_d \to C_{d-1}$ on d-simplices by

$$\delta_d(\sigma) = \sum_i (-1)^i [v_0, \ldots, \widehat{v_i}, \ldots v_d]_\sigma$$

and let it extend homomorphically to C_d. The alternating signs reflects the

ordering of the vertices in the simplex with respect to an underlying ordering of all the vertices:

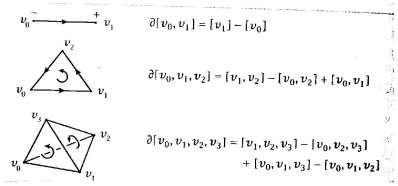


Figure 3: Figure from page 105 of [Ha].

For each $d \geq 0$ let $Z_d = Z_d(X) = \ker(\delta_d)$ and let $B_d = B_d(X) = \mathrm{im}(\delta_{d+1})$. We call the elements of these subgroups of C_d *d-cycles* and *d-boundaries* respectively.

An important point is that boundaries are cycles, so B_d is a subgroup of Z_i. Indeed this follows from the boundary condition:

Problem 4.2. Show that $\delta_{d+1} \circ \delta_d = 0$. That is, that $\delta_{d+1} \circ \delta_d(\sigma) = 0$ for any $(d+1)$-chain σ.

Thus we may define the dth homology group $H_d = H_d(X) = H_d^{\Delta}(X) = Z_d(X)/B_d(X)$, for all $d \geq 0$. We take δ_0 to be the null map $\delta : C_0 \to 0$, so $Z_0(X) = C_0(X)$.

4.3 An Example Computation of Homology Groups

Consider the simplicial complex structure on S^2 shown on the right, that has four 0-simplices $0, 1, 2$ and 3, six 1-simplices, four 2-simplices, and nothing else.

Let's calculate the homology. First off, we have,

$$\begin{aligned}
C_0 &= \langle [0], [1], [2], [3] \rangle = \mathbb{Z}^4 \\
C_1 &= \langle [0,1], [0,2], [0,3], [1,2], [1,3], [2,3] \rangle = \mathbb{Z}^6 \\
C_2 &= \langle [0,1,2], [0,1,3], [0,2,3], [1,2,3] \rangle = \mathbb{Z}^4 \\
C_d &= \emptyset \text{ for } d \geq 3
\end{aligned}$$

The boundaries are easy to calculate. By applying the boundary map to the basis of C_i we get a set that generates B_i. We then just have to find a independent subset of this generating set to get a basis of B_i, from which it is easy to

determine B_i.

$$\begin{aligned}
B_0 &= \delta_1(C_1) = \langle [1]-[0], [2]-[0], \ldots, [3]-[2] \rangle \\
&= \langle [1]-[0], [2]-[0], [3]-[0] \rangle \\
&= \mathbb{Z}^3 \\
B_1 &= \delta_2(C_2) = \langle [1,2]-[0,2]+[0,1], [1,3]-[0,3]+[1,0], \\
&\qquad [2,3]-[0,3]+[0,2], [2,3]-[1,3]+[1,2] \rangle \\
&= \mathbb{Z}^3 \text{ (the last one is the sum of the first three)} \\
B_d &= 0 \text{ for } d \geq 2.
\end{aligned}$$

Finding the cycles is a bit trickier, lets do the easy ones first. We have $Z_0 = C_0 \cong \mathbb{Z}^4$, so $H_0 = Z_0/B_0 \cong \langle [0] + B_0 \rangle = \mathbb{Z}$. For $d \geq 3$ we have that $Z_d = \delta_d(0) = 0$, so $H_d(X) = 0$.

We claim that $H_1 = 0$. Indeed, assume $z \in Z_1$ is non-trivial in Z_1/B_1. Let $\sum_i a_i\sigma_i + b$ for $b \in B_1$ be the element of the the coset $z + B_1$, (where each σ_i is from our basis of C_1,) that minimises $\sum_{i=1}^n |a_i|$. As $\delta(z) = 0$ and $\delta(b) = 0$ we have $\delta(\sum_i a_i\sigma_i) = 0$. As z is non-trivial, n is at least 1. Where $\sigma_1 = [v_1, v_2]$ there is some $\sigma_i = [v_2, v_3]$ (or $[v_3, v_2]$) with $a_i \neq 0$. Where $a = \min(|a_1|, |a_i|)$, we can, by inspecting B_1, replace $a\sigma_1 + a\sigma_i$ with $a[v_1, v_3]$ in $\sum_i a_i\sigma_i$ getting smaller $\sum_i |a_i|$.

We claim that $H_2 = \mathbb{Z}$. The chain $c_0 = [1,2,3] - [0,2,3] + [0,1,3] - [0,1,2]$ is in Z_2. We could apply δ to this to see this, or observe that if $[0,1,2,3]$ had been in C_3 then $\delta \circ \delta([0,1,2,3]) = 0$ by the boundary criteria, and $c_0 = \delta([0,1,2,3])$. As B_2 is empty, we have just to show c_0 generates Z_2. Consider any element $c = a_1[1,2,3] + a_2[0,2,3] + a_3[0,1,3] + a_4[0,1,2]$ in C_2. If $c \in Z_2$, then

$$\begin{aligned}
0 = \delta(c) &= (-a_3 - a_4)[0,1] \\
&+ (-a_2 + a_4)[0,2] \\
&+ \vdots \\
&+ (a_1 + a_2)[2,3]
\end{aligned}$$

One can check that this yields $a_1 = a_3 = -a_2 = -a_4$, so Z_2 is $\langle c_0 \rangle$.

In general, the hollow $d+1$-simplex, $\delta(\Delta^{d+1})$, is a simplicial complex structure for the sphere S^d. One can show much like we did above, that

$$H_i(\delta(\Delta^{d+1})) = \begin{cases} \mathbb{Z} & \text{if } i = 0 \text{ or } d \\ 0 & \text{otherwise} \end{cases}$$

We will also see that the simplicial homology groups are independent of choice of simplicial complex structure so we $H_i(S^d)$ is the same.

Problem 4.3. Give a simplicial complex structure to D^2 the bounded 2-dimensional disk. Compute the homology of this structure.

Problem 4.4. Calculate $H_i(S^2)$ using the Δ-complex structure given at the end of the last subsection.

Problem 4.5. Calculate the simplicial homology groups of the torus T, the projective plane $\mathbb{R}P^2$ and the Klein bottle K.

Hint:

Δ-complex structures are suggested on page 102 of [Ha].

Problem 4.6. Do exercise 9 from page 132 of [Ha]

5 Category Theory

A lot of proofs that we do for different homology theories, are basically the same. Category theory is s way to unify the proofs. For this section, we follow wikipedia: http://en.wikipedia.org/wiki/Category_theory (as of Jan 2011).

A *category* $\mathcal{C}$ consists of

- A set $\mathrm{Ob}(\mathcal{C})$ of *objects*.
- A set $\mathrm{morph}(X, Y)$ of *morphisms*, or *arrows* for each pair of objects $X, Y \in \mathrm{Ob}(\mathcal{C})$, including a distinguished identity morphism $\mathrm{id}_X \in \mathrm{morph}(X, X)$
- A composition of morphisms $\circ : \mathrm{morph}(X, Y) \times \mathrm{morph}(Y, Z) \to \mathrm{morph}(X, Z)$, taking $f \times g$ to $g \circ f$ which satisfies $f \circ \mathrm{id} = f = \mathrm{id} \circ f$ and $(h \circ g) \circ f = h \circ (g \circ f)$.

(This is actually a *small* category. More generally in a category, Ob and morph may be classes as opposed to just sets. I don't know the difference, but for the time being, it doesn't seem important.)

Problem 5.1. Show that for each $X \in \mathrm{Ob}(\mathcal{C})$, id_X is uniquely determined.

Problem 5.2. Given a category $\mathcal{C}$, show that $\mathcal{C}^{op}$ with the same set of objects, and with a distinct morphism f in $\mathrm{morph}_{\mathcal{C}^{op}}(Y, X)$ for every morphism f in $\mathrm{morph}_{\mathcal{C}}(X, Y)$, is also a category. Is it unique?

The categroy $\mathcal{C}^{op}$ is called the opposite category to $\mathcal{C}$. Many things that we say about $\mathcal{C}$, we get a similar statement about $\mathcal{C}^{op}$, but with the arrows reversed. Such a statement is the dual statement of the original statement. For any category cC let $\mathrm{op} : \mathcal{C} \to \mathcal{C}^{op}$ be the identity on objects, and take a morphism f in $\mathrm{morph}_{\mathcal{C}}(X, Y)$ to a morphism $\mathrm{morph}_{\mathcal{C}^{op}}(Y, X)$.

Some common categories are

- SET, whose objects are sets and whose morphisms are set maps
- GROUP, whose objects are group, and whose morphisms are homomorphism

- SPACE, whose objects are topological spaces, and whose morphisms are continuous functions,
- POINT, whose objects are topological spaces (X, x_0) with base point x_0 and whose morphisms are base-point fixing continuous funtions
- SIMP, whose objects are simplicial complices and whose morphisms are homomorphisms: vertex maps that take simplices to simplices.

A *(covariant) functor* $F : \mathcal{C} \to \mathcal{D}$ between categories assigns

- to each object X in $\mathcal{C}$, an object $F(X) \in \mathcal{D}$, and
- to each morphism $f : X \to Y$ in $\mathcal{C}$, a morphism $F(f) : F(X) \to F(Y)$ in $\mathcal{D}$,

such that the following properties hold.

- For each $X \in \mathrm{Ob}(\mathcal{C})$, $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$.
- For each $f \in \mathrm{morph}_{\mathcal{C}}(X, Y)$ and $g \in \mathrm{morph}_{\mathcal{C}}(Y, Z)$, $F(g \circ f) = F(g) \circ F(f)$.

A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ is the same except that it turns arrows around. That is, for each $f : X \to Y$ in $\mathcal{C}$ it assigns a morphism $F(f) : F(Y) \to F(X)$, and so the second property it must satisfy becomes $F(g \circ f) = F(f) \circ F(g)$.

Problem 5.3. Show that $\mathrm{op} : \mathcal{C} \to \mathcal{C}^{op}$ is a contravariant functor. Show that F is a covariant functor $F : \mathcal{C} \to \mathcal{D}$ if and only if $\mathrm{op} \circ F : \mathcal{C} \to \mathcal{D} \to \mathcal{D}^{op}$ is a covariant functor.

We say that a contravariant functor is *dual* to the corresponding covariant functor. Contravariant functors will come up more when we start looking at Hom modules and *cohomology*.

We have seen several functors already, all covariant. The fisrt was when we were looking at homotopy. Indeed, in Proposition 2.1 (i - iii) we showed that the map π taking the pointed space (X, x_0) in POINT to the group $\pi_1(X, x_0)$ and a morphism $\phi : (X, x_0) \to (Y, y_0)$ of pointed spaces to the homomorphism $\phi_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ satisfies the properties required of a functor.

In the same way, we could show that simplicial homology is a functor from the category DELTA of Δ-complexes structures (of spaces) to the category of abelian groups.

Problem 5.4. What should a morphism of Δ-complex structures be? Using this, show that the map $\Delta(X) \to H_i(\Delta(X))$ is a functor for each i.

Hint:

Δ-complex structures are, of course, spaces, so one could take continuous functions as morphisms. However, the point is to define morphisms that respect the Δ-complex structure of the objects.

A morphism $f : X \to Y$ in a category $\mathcal{C}$ is an *isomorphism* if there exists another morphism $g : Y \to X$ such that $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$.

Problem 5.5. Show that the isomorphisms in GROUP are group isomorphisms, and the isomorphisms in SPACE are homeomorphisms. What are isomorphisms in DELTA?

Property (iv) of Proposition 2.1 follows from the following statement about functors.

Proposition 5.1. *If $F : \mathcal{C} \to \mathcal{D}$ is a functor and $f \in \operatorname{morph}(\mathcal{C})$ is an isomorphisms, then $F(f) \in \operatorname{morph}(\mathcal{D})$ is an isomorphism.*

Hint:

Look at the proof of Proposition 2.1

Problem 5.6. Prove this.

Problem 5.7. Show that isomorphic Δ-complex structures give isomorphic homology groups.

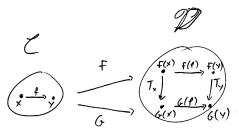
More is true, indeed Δ-complex structures that are homotopy equivalent (as spaces) give isomorphic homology groups. But this is not a property of the functor H_*^Δ.

Given a category such as the category of topological spaces, and two functors such as H_* and H_*^Δ we should like to know when these functors agree on arrows.

Definition 5.2. Let F and G be covariant functors from a category $\mathcal{C}$ to a category $\mathcal{D}$. Then a *natural transformation* T from F to G assigns a morphism $T_x : F(X) \to G(X)$ in $\mathcal{D}$ for each object X in $\mathcal{C}$ such that for each morphism $f : X \to Y$ in $\mathcal{C}$, we have

$$T_Y \circ F(f) = G(f) \circ T_X.$$

That is, the part of following diagram which is in $\mathcal{D}$, commutes.



6 Homological Algebra

The homological theory we develop applies for any abelian category, but as it is most common, we only do it for the category of modules over a given ring.

6.1 Modules

Recall the definition of a module.

Definition 6.1. Given a ring R with identity 1, an (left) *R-module* is an abelian group $(M, +)$ and an operation $R \times M \to M$ such that for all $r, s \in R$ and $m, n \in M$, we have

- $r(m + n) = rm + rn$
- $(r + s)m = rm + sm$
- $(rs)m = r(sm)$, and
- $1x = x$.

A *right R-module* is the same but the action of R is from the right: mr

Some examples of modules.

- A vector space over a field F is an F-module.
- A ring R is itself an R-module.
- A abelian group $(G, +)$ is a $\mathbb{Z}$-module: for z in $\mathbb{Z}$ the operation zg is $g + g + \cdots + g$ with z summands if z is positive.
- The 0 element of R is an R-module, called the 0-module. (In fact any subring of R is an R-module.)

There theory of modules is closely analogous to the theory of ableian groups.

A *submodule* of an R-module M is a subgroup closed under both operations: the group operation and action by R. A *homomorphism* $f : M \to N$ of R-modules is a homomorphism of groups with the additional property that $rf(m) = f(rm)$ for all $r \in R$ and $m \in M$. Sometimes to specify this additional property, we call such a homomorphism am *R-linear map*, or an *R-homomorphism*.

Problem 6.1. Show that a map $f : M \to N$ is an R-module homomorphism if and only if $f(rm + m') = r\phi(m) + f(m')$ for all $m, m' \in M$ and all r in R.

For a module M and a submodule N the factor-group M/N with elements $(m + N)$, is a module under action $r(m + N) = (rm + N)$. The quotient homomorphism of groups $M \to M/N$ is an R-homomorphism. For a homomorpism $f : M \to N$, the *kernel* $\ker(f)$ of f is the subgroup of M that maps to 0. The *image* $\operatorname{im}(f)$ of f is the subgroup $f(M)$ of N. These too are both modules. The *cokernal* $\operatorname{coker} f$ of f is the factor module $N/\operatorname{im}(f)$.

All the Isomorphism Theorems we know for groups have R-module analogues. In particular we have

Theorem 6.2 (The First isomorphism Theorem for Modules)**.** *Let $f : M \to N$ be an R-homomorphism. Then $M/ker f \cong f(M)$, and so $M/ker f \cong N$ if f is surjective.*

The proof is exactly the same as the proof for groups, but ground ourselves in the notation lets prove this.

Proof. Given an R-homomorphism $f : M \to N$, let $K = \ker f$ and $\phi : M \to M/K$ be the canonical factor homomorphism $\phi(m) = m+K$. Define $g : M/K \to N$ by $g(m+K) = f(m)$. If well defined, this clearly maps onto $f(M)$. We show that g is well defined, and an isomorphism.

To see that it is well defined, let $m' + K = m + K$, so $m' = m + k$ for some k with $f(k) = 0$. Then

$$g(m' + K) = f(m') = f(m + k) = f(m) + f(k) = f(m) = g(m + K).$$

To see that it is an R-homomorphism, observe $g(r(m + K) + (m' + K)) = g(rm + m' + K) = f(rm + m') = rf(m) + f(m') = rg(m + K) + g(m' + K)$.

For injectivity, assume that $g(m + K) = 0$. Then $f(m) = 0$ so $m \in K$. □

In proof of things we will start relying on commutative diagrams. Here is a starting point. Consider the commutative diagram:

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "u"] & C' \arrow[d, "v"] \\ D \arrow[r, "g"] & D' \end{tikzcd}$$

Proposition 6.3. *If the above diagram commutes, then*

- $u(\ker f) \subset \ker g$
- $v(\operatorname{im} f) \subset \operatorname{im} g$.

Proof. For $a \in \ker f$ we have $f(a) = 0$ and so

$$g \circ u(a) = v \circ f(a) = v(0) = 0,$$

which gives us that $u(a) \in \ker g$. For $b \in \operatorname{im} f$ there exists a in C with $f(a) = b$, so

$$v(b) = v \circ f(a) = g \circ u(a) = g(u(a)),$$

which gives us that $v(b) \in \operatorname{im} g$. □

It is thus immediate that u restricts to a homomorphism $u_* : \ker f \to \ker g$, and v restricts to a homomorphism $\operatorname{im} f \to \operatorname{im} g$.

The second item also gives us that v induces a homomorphism $v_* : \operatorname{coker} f \to \operatorname{coker} g$. Indeed $f(C) = \operatorname{im} f$ and $g(D) = \operatorname{im} g$, so the second item gives us that the factor map

$$v_* : C'/f(C) \to D'/g(D) : b + f(C) \mapsto v(b) + g(D)$$

is well defined map between the cokernels. It is clearly a homomorphism.

Problem 6.2. Show that the diagram

$$\begin{tikzcd} C' \arrow[r, "\phi"] \arrow[d, "v"] & C'/f(C) \arrow[d, "v_*"] \\ D' \arrow[r, "\phi"] & D'/g(D) \end{tikzcd}$$

also commutes, where the ϕ are the canonical factor maps, and v_* is the induced map from above.

The direct sum and product for modules are also defined in the obvious analogue to their definitions for groups. In particular we have.

Definition 6.4. Let $(M_i)_I$ be a family of R-modules, then the direct sum $\bigoplus_{i \in I} M_i$ is the set of tuples $(m_i)_I$ with $m_i \in M_i$ for all i and all but finitely many are 0. The action of R on $\bigoplus M_i$ is componentwise: $r(m_i)_I = (rm_i)_I$.

One important derived module that we will see is the module of R-module homomorphisms. For r-modules M and N let $\operatorname{Hom}_R(M, N)$ be the set of R-module homomorphisms from M to N. For homomorphisms $f, g \in \operatorname{Hom}_R(M, N)$, let $f + g$ be defined $(f + g)(m) = f(m) + g(m)$ for all $m \in M$. This is also a homomorphism:

$$\begin{aligned} (f + g)(rm + m') &= f(rm + m') + g(rm + m') \\ &= rf(m) + f(m') + rg(m) + g(m') \\ &= r(f + g)(m) + (f + g)(m') \end{aligned}$$

Problem 6.3. Show that $\operatorname{Hom}_R(M, N)$ is an abelian group under this operation. Show that if R is commutative, $\operatorname{Hom}_R(M, N)$ is an R-module with R-action defined by $(rf)(m) = f(rm)$.

We speak of an arbitrary fixed ring R, so will say *module* to mean R-module.

6.2 Exact Sequences

A sequence

$$\dots \xrightarrow{f_{n+2}} M_{n+1} \xrightarrow{f_{n+1}} M_n \xrightarrow{f_n} M_{n-1} \xrightarrow{f_{n-1}} M_{n-2} \xrightarrow{f_{n-2}} \dots$$

where $f_i : M_{i+1} \to M_i$ is a homomorphism of modules, is *exact at* M_i if $\operatorname{im} f_{i+1} = \ker f_i$. The sequence is *exact* if it is exact at each M_i.

This definition applies to a finite sequence such as

$$M_i \to M_{i-1} \to \cdots \to M_1 \to M_0$$

by extending it with the 0 module:

$$\dots 0 \to 0 \to M_i \to M_{i-1} \to \cdots \to M_1 \to M_0 \to 0 \to 0 \to \dots$$

This is clearly exact at each group 0.

Problem 6.4. Show that g is injective if and only if $0 \to B \xrightarrow{g} C$ is exact, and that f is surjective if and only if $A \xrightarrow{f} B \to 0$ is exact.

A *short sequence* is a sequence $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$. If it is exact, then we can say the following.

- Exactness at A gives us that f is injective, so allows us to view A as the submodule $\operatorname{im} f$ of B. Thus $\operatorname{coker} f \cong B/A$.
- Exactness at C gives us that g is surjective, so by the first isomorphism theorem for modules, we have that $C \cong B/\ker g$.
- Exactness at B gives us that $\ker g = \operatorname{im} f = A$, so $C \cong B/A \cong \operatorname{coker} f$.

Problem 6.5. Do problem 14 from page 132 of [Ha].

Now consider the following commuting diagram in which the columns are exact.

$$\begin{tikzcd}
 & 0 \arrow[d] \\
A \arrow[r, "\alpha"] \arrow[d, "f"] & A' \arrow[d, "f'"] \\
B \arrow[r, "\beta"] \arrow[d, "g"] & B' \arrow[d, "g'"] \\
C \arrow[r, "\gamma"] \arrow[d] & C' \\
0 &
\end{tikzcd}$$

For an element z in $\ker \gamma$ we define a element $d(z)$ in $\operatorname{coker} \alpha$ as follows. By exactness at C, g is surjective, so there is y in $g^{-1}(z)$. By the commutativity of the bottom square, $g' \circ \beta(y) = \gamma \circ g(y) = 0$, so $\beta(y) \in \ker g'$. By exactness at B', $\beta(y)$ is in $\operatorname{im} f'$ and by exactness at A', f' is invertible on its image, so let $d(z) = {f'}^{-1} \circ \beta \circ g^{-1}(z)$ is unique up to the choice of $y \in g^{-1}(z)$.

For another choice y' in $g^{-1}(z)$, there is, by exactness at B, some x in A such that $f(x) = (y - y')$. By the commutativity of the top square $f' \circ \alpha(x) = \beta \circ f(x) = \beta(y - y') = \beta(y) - \beta(y')$. As f' is invertible on its image, we have that ${f'}^{-1} \circ \beta(y) - {f'}^{-1} \circ \beta(y') = \alpha(x) \in coker(\alpha)$. Thus $d : \ker \gamma \to \operatorname{coker} \alpha$ is well defined.

For a more dramatic, and concise version of the above discussion see the clip http://www.youtube.com/watch?v=etbcKWEKnvg from the movie "It's My Turn".

The following Lemma is frighteningly referred to as the Snake Lemma.

Lemma 6.5. *If the diagram above commutes, and the columns are exact, then for $z \in \ker \gamma$, $d(z) = {f'}^{-1} \circ \beta \circ g^{-1}(z)$ is well defined, and the following sequence is exact:*

$$\ker \alpha \longrightarrow \ker \beta \longrightarrow \ker \gamma \xrightarrow{d} \operatorname{coker} \alpha \longrightarrow \operatorname{coker} \beta \longrightarrow \operatorname{coker} \gamma$$

The unlabeled maps, from left to right, are the natural maps induced by f, g, f' and g'.

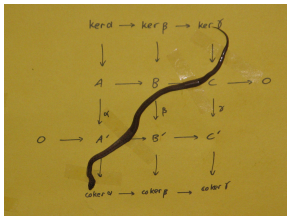
Proof. We showed that d is well-defined prior to stating the lemma. To finish the proof, we must show that the sequence is exact.

We show that it is exact at $\ker\beta$. We must show that $\operatorname{im} f_* = \ker g_*$, where recall that $f_* : \ker\alpha \to \ker\beta : x \to f(x))$ and $g_* : \ker\beta \to \ker\gamma$. Clearly $g_* \circ f_* = (g \circ f)_* = 0_* = 0$ so we have $\operatorname{im} f_* \subset \ker g_*$. To see that $\ker g_* \subset \operatorname{im} f_*$, let $x \in \ker g_*$. By exactness at B there is $y \in A$ such that $f(y) = x$. To show that $x \in \operatorname{im} f_*$, as we need, we show that $y \in \ker\alpha$. Indeed, by the commutativity of the upper square, $f' \circ \alpha(y) = \beta \circ f(y) = \beta(x) = 0$. As f' is injective, (by exactness at A'), $\alpha(y) = 0$, and so $y \in \ker\alpha$.

The other parts are exercises. $\square$

Problem 6.6. Pick a module in the sequence in Lemma 6.5, and show that the sequence is exact there. Don't do the same one as your buddy.

This picture from http://concretenonsense.wordpress.com/2010/04/19/a-taxidermic-snake-lemma/ explains the name of the lemma.



6.3 Chain Complexes and Homology

A chain complex C is a sequence

$$\dots \xrightarrow{\delta_{i+1}} C_i \xrightarrow{\delta_i} C_{i-1} \xrightarrow{\delta_{i-1}} \dots \xrightarrow{\delta_2} C_1 \xrightarrow{\delta_1} C_0 \xrightarrow{\delta_0} 0,$$

of modules $C_0, C_1, C_2, \dots$ connected by homomorphisms $\delta_i : C_i \to C_{i-1}$, which we call boundary homorpishms, such that for all i,

$$\delta_i \circ \delta_{i+1} = 0$$

This condition is equivalent to the condition that $\operatorname{im}\delta_{i+1} \subseteq \ker\delta_i$.

Elements of C_i are called i-chains. For each i let $B_i = \operatorname{im}\delta_{i+1}$ and $Z_i = \ker\delta_i$. The elements of these submodules of C_i are called the i-boundaries and i-cycles respectively. As $\operatorname{im}\delta_{i+1} \subseteq \ker\delta_i$, B_i is a submodule of Z_i. Let

$$H_i(C) = H_i = Z_i/B_i = \ker\delta_i/\operatorname{im}\delta_{i+1}.$$

This module, a submodule of $\operatorname{coker}\delta_{i+1}$, is the ith homology module of the chain complex C.

Often we will have families of maps such as the family $\{\delta_i\}$ of boundary maps defined above. We will often refer to them collectively as $\delta = \{\delta_i\}$, and where it causes no confusion, drop the index in referring to individual maps. For example, we can denote the condition that $\delta_i \circ \delta_{i+1} = 0$ for all i by saying that $\delta \circ \delta = 0$. The understanding is that it holds for every choice of indices for which it makes sense.

Taking the modules as abelian groups (so $\mathbb{Z}$-modules) this applies with everything being a group. Compare it to our simplicial homology.

Example 6.6. Let C be the chain complex of abelian groups defined by

- $C_0 = C_1 = \mathbb{Z}$ and C_i is otherwise 0
- δ_1 is 2, (multiplication by 2).

The other boundary maps are uniquely determined. We compute the homology groups of C.

The kernel Z_0 of δ_0 is $C_0 = \mathbb{Z}$, and the image B_0 of $\delta_1 = 2$ is $2\mathbb{Z}$, so $H_0(C) = \mathbb{Z}/2\mathbb{Z}$, the group on two elements.

The kernel Z_1 of δ_1 is 0, and so $H_1(C)$ is 0. For $i \geq 2$, $Z_i \subset B_i = 0$ so $H_i(C) = 0$.

Observe that if the chain complex C is exact, then the homology modules are all trivial. So homology measures the failure of the sequence to be exact.

6.4 Homotopy Invariance

A morphism $u : C \to D$ of chain complexes, or a *chain map*, is a family of module homomorphisms $u_i : C_i \to D_i$ such that the following commutes.

$$\begin{tikzcd}
\cdots \arrow[r] & C_2 \arrow[r, "\delta_2"] \arrow[d, "u_2"] & C_1 \arrow[r, "\delta_1"] \arrow[d, "u_1"] & C_0 \arrow[r, "\delta_0"] \arrow[d, "u_0"] & 0 \\
\cdots \arrow[r] & D_2 \arrow[r, "\delta_2"] & D_1 \arrow[r, "\delta_1"] & D_0 \arrow[r, "\delta_0"] & 0
\end{tikzcd}$$

We write the commutativity of this diagram succinctly as $\delta \circ u = u \circ \delta$, meaning that this holds for any choice of indices for which it makes sense.

Applying Proposition 6.3 to this diagram with f and g as the two boundary maps δ_{i+1} we get that u_i induces a map $u_{*,i} : B_i(C) \to B_i(D)$ of the i-boundaries of C to the i-boundaries of D. Applying it with f and g as the maps δ_i, we get that u_i induces a map $u_{*,i} : Z_i(C) \to Z_i(D)$ of i-cycles. And so the map on boundaries descends to a homomorphism $u_{*,i} : H_i(C) \to H_i(D)$ of the homology modules.

We will look at conditions on such a chain map u that imply that the induced homomorphism u_* is an isomorphism. This will help us show the relation between simplicial and singular homology.

As a first step, we look at a condition on a pair $u, v : C \to D$ of chain maps that implies $u_* = v_*$.

Definition 6.7. Given a pair $u, v : C \to D$ of chain maps from the complex C to the complex D, a *chain homotopy* h between u and v is collection $h = \{h_i\}_{i \geq -1}$ of module homomorphisms $h_i : C_i \to D_{i+1}$ (with $h_{-1} : 0 \to D_0$) such that $\delta \circ h + h \circ \delta = v - u$. That is, for each i, we have the following equality of maps on C_i:

$$\delta_{i+1} \circ h_i + h_{i-1} \circ \delta_i = v_i - u_i.$$

If there is a chain homotopy between u and v, we say that they are *chain homotopic*.

This is visualised by the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & C_2 \arrow[r, "\delta_2"] \arrow[dl, "h_3"'] \arrow[d, shift right, "u_2"'] \arrow[d, shift left, "v_2"] & C_1 \arrow[r, "\delta_1"] \arrow[dl, "h_2"'] \arrow[d, shift right, "u_1"'] \arrow[d, shift left, "v_1"] & C_0 \arrow[r, "\delta_0"] \arrow[dl, "h_1"'] \arrow[d, shift right, "u_0"'] \arrow[d, shift left, "v_0"] & 0 \arrow[dl, "h_0"'] \\
\cdots \arrow[r] & D_2 \arrow[r, "\delta_2"] & D_1 \arrow[r, "\delta_1"] & D_0 \arrow[r, "\delta_0"] & 0
\end{tikzcd}$$

Proposition 6.8. *Given a chain homotopy h of chain maps $u, v : C \to D$ from the complex C to the complex D, the induced maps $u_*, v_* : H_*(C) \to H_*(D)$ on homology are equal.*

Proof. Fixing an index $i \geq 0$, we want to show that for $[x] \in H_i(C)$, we have $u_*([x]) = v_*([x])$ in $H_i(D)$. That is, for some i-cycle x in $Z_i(C)$, that $[u(x)] = [v(x)]$, or equivalently that $v(x) - u(x) \in B_i(D)$.

Now, we have that

$$v(x) - u(x) = (v - u)(x) = \delta_{i+1} \circ h_i(x) + h_{i-1} \circ \delta_i(x).$$

Since $B_i(D) = \operatorname{im} \delta_{i+1}$ by definition, $\delta_{i+1} \circ h_i(x) \in B_i(D)$, and since $x \in Z_i(C)$, $\delta_i(x) = 0$, so $h_{i-1} \circ \delta_i(x) = 0 \in B_i(D)$. Thus the sum is in $B_i(D)$, as needed. $\square$

In analogue to homotopy equivalence of spaces, we have the following for chains.

Definition 6.9. A chain map $u : C \to D$ is a *chain homotopy equivalence* if there is a chain map $v : D \to C$ such that $v \circ u$ and $u \circ v$ are chain homotopic to the identity maps on C and D respectively.

Proposition 6.10. *If $u : C \to D$ is a* chain homotopy equivalence *then $u_* : H_*(C) \to H_*(D)$ is an isomorphism.*

Problem 6.7. Prove this.

The choice of the name *chain homotopy equivalence* reflect the fact that for homotopically equivalent spaces, we will be able to define a chain homotopy equivalence between their complexes of chains, and so they will have isomorphic homology. This is a little difficult to show directly for the simplicial homology. So we will define another version of homology which makes this easier to prove.

6.5 The Long Exact Sequence of Homology

A short exact sequence

$$0 \longrightarrow A \xrightarrow{u} B \xrightarrow{v} C \longrightarrow 0$$

of chain maps is exact if it is exact for every index, that is, if each column of the following diagram (which commutes) is exact:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & A_{i+1} \arrow[r, "\delta"] \arrow[d, "u"] & A_i \arrow[r, "\delta"] \arrow[d, "u"] & A_{i-1} \arrow[r] \arrow[d, "u"] & \cdots \\
\cdots \arrow[r] & B_{i+1} \arrow[r, "\delta"] \arrow[d, "v"] & B_i \arrow[r, "\delta"] \arrow[d, "v"] & B_{i-1} \arrow[r] \arrow[d, "v"] & \cdots \\
\cdots \arrow[r] & C_{i+1} \arrow[r, "\delta"] \arrow[d] & C_i \arrow[r, "\delta"] \arrow[d] & C_{i-1} \arrow[r] \arrow[d] & \cdots \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

We have for each i, that the maps u and v induce maps short sequences

$$H_i(A) \xrightarrow{u_*} H_i(B) \xrightarrow{v_*} H_i(C).$$

Further, the snake lemma gives us a map $d_i : \ker \delta \to \operatorname{coker} \delta$ which is well defined by $d_i(z) = u^{-1} \circ \delta \circ v^{-1}(z)$. In fact, d_i maps to the submodule $H_{i-1}(A)$ of $\operatorname{coker}(\delta)$. To see this we must show that $d_i(z) \in \ker(\delta)$.

$\ker \delta \subset A_{i-1}$. Indeed where $u \circ \delta(d_i(z)) = \delta \circ u(d_i(z))$ which is $\delta(\delta \circ v^{-1}(z)) = 0$ as B is a chain complex. By the exactness of the columns u is injective, so $\delta(d_i(z)) = 0$.

Usually the map d_i is denoted δ_i and called the *connecting map*. Thus the short exact sequence $0 \longrightarrow A \xrightarrow{u} B \xrightarrow{v} C \longrightarrow 0$ of chain maps induces a long sequence

$$\cdots \xrightarrow{v_*} H_{i+1}(C) \xrightarrow{\delta} H_i(A) \xrightarrow{u_*} H_i(B) \xrightarrow{v_*} H_i(C) \xrightarrow{\delta} H_{i-1}(A) \xrightarrow{u_*} \cdots$$

on homology modules.

In fact we have the following.

Theorem 6.11. *The sequence*

$$\cdots \xrightarrow{v_*} H_{i+1}(C) \xrightarrow{d} H_i(A) \xrightarrow{u_*} H_i(B) \xrightarrow{v_*} H_i(C) \xrightarrow{\delta} H_{i-1}(A) \xrightarrow{u_*} \cdots$$

is exact.

Problem 6.8. Using the Snake Lemma, prove the theorem.

The long exact sequence of homology groups provides us examples of a natural transformations: from the category of short exact sequences of chain complexes to the category of modules. Indeed for each module in the long exact sequence can be viewed as a functor. For example we may view $H_i(B)$ as the image of a short exact sequence $0 \to A \to B \to C \to 0$ under the functor $H_{i,2}$, where the subscript $i, 2$ denotes that it is the ith homology group of the 2nd of the three chain compexes in the sequence. The induced map $v_* : H_i(B) \to H_i(C)$ can be viewed as a transformation between the functors $H_{i,2}$ and $H_{i,3}$. And indeed, it is natural. All such transformations are natural.

This is a bit too much category theory. What it means in dirty terms, is that for a morphism

$$f : (0 \to A \to B \to C \to 0) \longrightarrow (0 \to A' \to B' \to C' \to 0)$$

of short exact sequences of chain complexes (which is a chain map f_i for each short exact sequence in the complex, that commutes with boundary maps, see [Ha] page 127), the following induced diagram commutes.

$$\begin{tikzcd}
\cdots \arrow[r] & H_i(A) \arrow[r, "u_*"] \arrow[d, "f_*"] & H_i(B) \arrow[r, "v_*"] \arrow[d, "f_*"] & H_i(C) \arrow[r, "\delta"] \arrow[d, "f_*"] & H_{i-1}(A) \arrow[r] \arrow[d, "f_*"] & \cdots \\
\cdots \arrow[r] & H_i(A') \arrow[r, "u_*"] & H_i(B') \arrow[r, "v_*"] & H_i(C') \arrow[r, "\delta"] & H_{i-1}(A') \arrow[r] & \cdots
\end{tikzcd}$$

Problem 6.9. Prove this.

Now we could continue to develop the theory of Homology algebraically. But let's move back to topology again, and apply some of ideas we have just seen.

7 Singular Homology

We saw that the Homological Algebra we developed generalises Simplicial Homology of topological spaces. It can be applied to various Homology Theories. Singular Homology is another such theory for topological spaces.

7.1 Singular Homology

In simplicial homology we defined the d-chains C_d as the free group, or $\mathbb{Z}$-module, of d-simplices in a simplicial complex structure Δ on a space X. In singular homology, we replace d-simplices of a structure Δ with all singular d-simplices in X.

A *singular d-simplex* in a space X is a continuous function

$$\sigma : \Delta^d \to X,$$

where Δ^d is the standard d dimensional simplex. Where $e_0, \ldots, e_d$ are the standard vertices of Δ^d and $v_i = \sigma(e_i)$ for $i = 0, \ldots, d$, we often write σ as

$$[v_0, v_1, \ldots, v_d]_\sigma,$$

dropping even the subscript if it is clear. The image of the face of Δ^d which omits the vertex e_i is denoted $[v_0, \ldots, \widehat{v_i}, \ldots, v_d]_\sigma$ and called a face of $[v_0, v_1, \ldots, v_d]_\sigma$. Let C_d be the free group (free $\mathbb{Z}$-module) generated by the singular d-simplices of X.

Define the *boundary* map $\delta_d : C_d \to C_{d-1}$ by setting $\delta_d(\sigma)$ for a singular d-simplex σ of X to be the sum

$$\delta_d(\sigma) = \sum_{k=0}^{d} (-1)^k [v_0, \ldots, \widehat{v_k}, \ldots, v_d]_\sigma$$

of σ restricted to the faces (boundary $(d-1)$-simplices) of Δ^d. Extend δ_d to C_d homomorphically.

One can verify that this defined a chain complex C. The corresponding homology groups $H_i(X) = H_i(C)$ are called the *singular homology groups of X*.

Problem 7.1. Show that for a point X, the homology group $H_0(X)$ is $\mathbb{Z}$, and $H_i(X)$ is 0 for $i \geq n$.

Problem 7.2. Show that $H_0(X) \cong \mathbb{Z}$ for any path connected space X.

Problem 7.3. Replace the chain complex

$$\cdots \to C_2(X) \xrightarrow{\delta_2} C_1(X) \xrightarrow{\delta_1} C_0(X) \xrightarrow{0}$$

which you used in the previous question with

$$\cdots \to C_2(X) \xrightarrow{\delta_2} C_1(X) \xrightarrow{\delta_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \xrightarrow{0}$$

where $\varepsilon(\sigma) = 1$ for any singular 0-simplex σ. Compute the homology groups $\widetilde{H}_i(X) = H_i(C)$ of this chain complex C.

The *augmented chain complex*

$$\widetilde{C} = \cdots \to C_2 \xrightarrow{\delta_2} C_1 \xrightarrow{\delta_1} C_0 \xrightarrow{\varepsilon} \mathbb{Z} \xrightarrow{0}$$

is defined in the same way for any chain complex C. The homology $\widetilde{H}_*(C) = H_*\widetilde{C}$ of this augmented complex is called the *reduced homology of* C. It can be viewed as a normalisation of the homology of C.

Problem 7.4. Show that for any chain complex C, $\widetilde{H}_i(C)$ is $H_i(C)$ if $i \geq 1$ and $H_i(C) \oplus \mathbb{Z}$ if $i = 0$.

Problem 7.5. Show that where $X_1, \ldots, X_c$ are the path connected components of a space X, that

$$H_i(X) \cong \bigoplus_{\alpha=1}^{c} H_i(X_\alpha)$$

for all i.

7.2 Homotopy Equivalence

In singular homology, our chains groups are not generally finitely generated like they are in simplicial homology, so the singular homology groups $H_*(X)$ of a space X are perhaps harder to compute than the simplicial homology groups $H_*^\Delta(X)$. However, their 'more topological' definition makes proving homological properties easier. Further, we will show that in the case that X is a Δ-complex, $H_*^\Delta(X) \cong H_*(X)$, which means these properties will be proved for simplicial homology as well.

In particular it is clear that a continuous function $f : X \to Y$ of spaces induces a set of homomorphisms $f_\sharp : C_d(X) \to C_d(Y)$ defined on singular d-simplices by $f_\sharp\sigma = f \circ \sigma$. We see that $f_\sharp$ is a chain map:

$$\begin{aligned} f_\sharp \circ \delta(\sigma) &= f_\sharp \sum_{k=0}^{d} (-1)^k [v_0, \ldots, \widehat{v_k}, \ldots, v_d]_\sigma \\ &= \sum_{k=0}^{d} (-1)^k [v_0, \ldots, \widehat{v_k}, \ldots, v_d]_{f\circ\sigma} \\ &= \delta(f \circ \sigma) = \delta \circ f_\sharp(\sigma) \end{aligned}$$

So $f_\sharp$ induces a homomorphism $f_* : H_d(X) \to H_d(Y)$ of homology groups.

Theorem 7.1. *If $f, g : X \to Y$ are homotopic continuous functions, then they induce the same homomorphism $f_* = g_* : H_d(X) \to H_d(Y)$ of homology groups.*

Proof. We use the homotopy $H : X \times I \to Y$ to define a chain homotopy h between $f_\sharp, g_\sharp : C(X) \to C(Y)$. The theorem then follows from Proposition 6.8.

We need to define a map h_d on singular d-simplices of X to singular $(d+1)$-simplices of Y, such that $g_\sharp - f_\sharp = h_{d-1} \circ \delta_d + \delta_{d+1} \circ h_d$. Such a map extends uniquely to a homomorphism of the chain groups, and this homomorphism satisfies the same identity.

We begin by defining the *prism* decomposition– a decomposition of the prism $\Delta^d \times I$ into $d+1$ different $(d+1)$-simplices $p_0, \ldots, p_d$. For $i = 0, \ldots, d$, let a_i and b_i be the vertices $(e_i, 0)$ and $(e_i, 1)$ respectively, of $\Delta^d \times I$. Let p_i be the $(d+1)$-simplex $[a_0, a_1, \ldots, a_i, b_i, b_{i+1}, \ldots b_d]$. One can check that this is in fact a simplicial complex structure on $\Delta^d \times I$.

Now for a singular d-simplex $\sigma : \Delta^d \to X$, let H_σ be the composition

$$H_\sigma = H \circ (\sigma \times \mathrm{id}) : \Delta^d \times I \to X \times I \to Y.$$

So H_σ restricted to the $(d+1)$-simplex p_i is in $C_{d+1}(Y)$ and is represented $[a_0, a_1, \ldots, a_i, b_i, b_{i+1}, \ldots b_d]_{H_\sigma}$ where $f_\sharp\sigma = [a_0, \ldots a_d]_{f_\sharp\sigma}$ and $g_\sharp\sigma = [b_0, \ldots b_d]_{g_\sharp\sigma}$.

Let

$$h_d(\sigma) = \sum_{i=1}^{d} (-1)^i [a_0, a_1, \ldots, a_i, b_i, b_{i+1}, \ldots b_d]_{H_\sigma}.$$

Then to verify that $g_\sharp - f_\sharp = h_{d-1} \circ \delta_d + \delta_{d+1} \circ h_d$, observe that

$$\begin{aligned} \delta_{d+1} \circ h_d(\sigma) &= \delta_{d+1} \sum_{i=0}^{d} (-1)^i [a_0, \ldots, a_i, b_i, \ldots, b_d] && (1) \\ &= \sum_{j \leq i} (-1)^{i+j} [a_0, \ldots, \widehat{a_j}, \ldots, a_i, b_i, \ldots, b_d] && (2) \\ &+ \sum_{j > i} (-1)^{i+j} [a_0, \ldots, a_i, b_i, \ldots, \widehat{b_{j-1}}, \ldots, b_d]. && (3) \end{aligned}$$

Now for $\alpha = 1, \ldots d-1$ the $i = j = \alpha$ term

$$(-1)^{2\alpha} [a_0, \ldots, a_{\alpha-1}, \widehat{a_\alpha}, b_\alpha, \ldots, b_d]$$

in (2) cancels with the $i = j - 1 = \alpha - 1$ term

$$(-1)^{2\alpha-1} [a_0, \ldots, a_{\alpha-1} \widehat{b_{\alpha-1}}, b_\alpha, \ldots, b_d],$$

in (3). This leaves the the $i = j = 0$ term

$$[\widehat{a_0}, b_0, \ldots b_d] = g_\sharp\sigma,$$

the $i = j - 1 = d$ term

$$(-1)^{2d+1}[a_0, \dots a_d, \widehat{b_d}] = -f_\sharp \sigma,$$

and the terms

$$\begin{aligned}
& \sum_{j<i} (-1)^{i+j}[a_0, \dots, \widehat{a_j}, \dots, a_i, b_i, \dots, b_d] \\
+ & \sum_{j>i+1} (-1)^{i+1+j}[a_0, \dots, a_i, b_i, \dots, \widehat{b_{j+1}}, \dots, b_d] \\
= & -\sum_{i \geq j} (-1)^{i+j}[a_0, \dots, \widehat{a_j}, \dots, a_{i+1}, b_{i+1}, \dots, b_d] \\
+ & -\sum_{i<j} (-1)^{i+j}[a_0, \dots, a_i, b_i, \dots, \widehat{b_j}, \dots, b_d] \\
= & -h_{d-1} \sum_{j=0}^{d} (-1)^j [v_0, \dots, \widehat{v_j}, \dots, v_d] \\
= & -h_{d-1} \circ \delta_d(\sigma)
\end{aligned}$$

We thus have that $\delta \circ h = g_\sharp - f_\sharp - h \circ \delta$, as needed. □

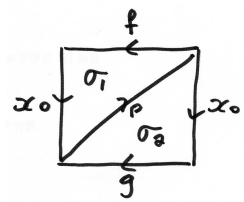
Corollary 7.2. *If $f : X \to Y$ is a homotopy equivalence, then $f_* : H_d(X) \to H_d(Y)$ is an isomorphism.*

7.3 Homology and the Fundamental Group

As loops $f : I \to X$ in a space X can be viewed as singular one cycles, there is a natural map h from $\pi_1(X)$ to $H_1(X)$. We show more.

Theorem 7.3. *For a space H is a homomorphism $h : \pi_1(X, x_0) \to H_1(X)$. If X is path connected, it is surjective, and its kernal is the commutator subgroup π_{ab} of $\pi_1(X, x_0)$.*

Proof. For (the class of) a loop $f \in \pi_1(X, x_0)$ let $h(f) = f$. To see that this is well defined, we must show that homotopic loops $f \equiv g$, when viewed as 1-simplices, are also homologous $f \sim g$ (that it, that $f - g$ is the boundary of a 2-chain in X). This is easy. Splitting the homotopy of paths from g to f into two 2-cycles with the path p (x_0 is the constant path at x_0):

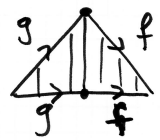


we get that $\delta(\sigma_1 + \sigma_2) = f + x_0 + p - g - x_0 - p = f - g$, as needed. So $h : f \mapsto f$, where the first f is the homotopy class of f, and the second is the homology class of f, is well defined.

To see that it is a homomorphism we have to show that it preserves operations: identity, inverse, and the group operation ($\cdot$ in $\pi_1(X, x_0)$ and $+$ in $H_1(X)$).

The identity x_0 in $\pi_1(X, x_0)$ maps to the class in $H_1(X)$ of the constant 1-simplex $[x_0, x_0]$. This is the boundary of the constant 2-simplex $[x_0, x_0, x_0]$ (indeed $\delta([x_0, x_0, x_0]) = [x_0, x_0] - [x_0, x_0] + [x_0, x_0] = [x_0, x_0]$,) so is the zero class, or identity element of $H_1(X)$.

The boundary of the 2-simplex below (which is constant on vertical lines) is $g + f - g \cdot f$ so $g \cdot f$ in $\pi_1(X)$ maps to the class of $g + f$ in $H_1(X)$. So h preserves the group operation.



As h preserves identity and concatination, it preserves inverse, taking f^{-1}, the reverse of the loop f, to $-f$. So we have shown that it is a homomorphism.

Let X be path connected. To see that h is surjective (onto $H_1(X)$), let σ be an element of $Z_1(X)$. As σ is a cycle, it follows from Euler's theorem that σ can be written as a sum of (graph) cycles : sums of the form $c = \sum_{i=1}^{d} [v_d, v_{d+1}]$ where indices are modulo d (so $v_{d+1} = v_1$). Viewing a 1-simplex as a path in X, $[x_0, v_1] \cdot [v_1, v_2] \cdot \dots \cdot [v_d, v_1] \cdot [v_1, x_0]$ is loop at x_0 in X whose image under h is $[x_0, v_1] + c - [x_0, v_1] = c$. As such cycles c are in the image of h, their sum σ is also in the image of h.

We now check that the kernal of h is the commutator subgroup π_{ab} (generated by all elements of the form $f \cdot g \cdot f^{-1} \cdot g^{-1}$) of $\pi_1(X, x_0)$. It certainly contains π_{ab}, as $H_1(X)$ is commutative. Let $h(f) = 0$ for some loop $f \in \pi_1(X, x_0)$. Then $h(f) = \sum_i \delta(\sigma_i)$ for some multi-set of 2-simplices σ_i.

Let $\sigma_i = [u_i, v_i, w_i]$ and let $\delta(\sigma_i)$ denote the closed path $[v_i, w_i] \cdot [u_i, w_i]^{-1} \cdot [u_i, v_i]$ in X (so that $h(\delta\sigma_i) = \delta\sigma_i$ and $f = \sum_i \delta\sigma_i$). For each point $v \in X$ let p_v be a path from x_0 to v.

For each i, let τ_i be the loop

$$\tau_i = p_{v_i} \cdot [v_i, w_i] \cdot p_{w_i}^{-1} \cdot p_{w_i} \cdot [u_i, v_i]^{-1} \cdot p_{u_i}^{-1} \cdot p_{u_i} \cdot [u_i, v_i] \cdot p_{v_i}^{-1}.$$

Clearly, we have

$$\tau_i \simeq p_{v_i} \cdot \delta(\sigma_i) \cdot p_{v_i}^{-1}, \tag{4}$$

which is homotopic to x_0. So $\sum_i \tau_i \simeq 0$ in $\pi_1(X, x_0)$.

Further, passing to $\pi_1(X, x_0)/\pi_{ab}$, we have that

$$\sum_i \tau_i = \sum_i p_{v_i} \cdot \delta(\sigma_i) \cdot p_{v_i}^{-1} \equiv \sum_i \delta(\sigma_i) = f.$$

Since $\tau_i \simeq x_0$ in $\pi_1(X, x_0)$ and so $\tau_i \equiv x_0$ modulo π_{ab}, we get that $f \equiv \sum_i \tau_i$ is in π_{ab}. This is what we had to show. $\square$

Problem 7.6. Show that τ_i from (4) is indeed homotopic to the constant loop x_0.

7.4 Relative Homology

For a subspace A of a space X, any singular i-simplex of A is also a singular i-simplex of X. So $C_i(A)$ is a submodule of $C_i(X)$. Clearly the diagram on the right, where the horizontal maps are injections, commutes. According to Problem 6.2, it therefore induces the commuting diagram below.

$$\begin{tikzcd}
C_i(A) \arrow[r, "\text{inj}"] \arrow[d, "\delta"] & C_i(X) \arrow[d, "\delta"] \\
C_{i-1}(A) \arrow[r, "\text{inj}"] & C_{i-1}(X)
\end{tikzcd}$$

$$\begin{tikzcd}
C_i(X) \arrow[r, "\text{quotient}"] \arrow[d, "\delta"] & C_i(X)/C_i(A) \arrow[d, "\delta_*"] \\
C_{i-1}(X) \arrow[r, "\text{quotient}"] & C_{i-1}(X)/C_{i-1}(A)
\end{tikzcd}$$

Let $C_i(X, A)$ be the quotient module $C_i(X)/C_i(A)$ and denote the induced boundary map δ_* simply by δ, clearly $\delta \circ \delta = 0$. Thus $C(X, A)$ is a chain complex. Its homology $H_*(X, A)$ is called the *homology of X relative to (or modulo) A*.

Problem 7.7. Define substructure relations for chain complexes. For a subchain complex C' of C define a quotient chain complex C/C'. Use this to define a sequence of homology groups of C relative to C' which generalise the Homology of a space X relative to a subspace A.

Consider the module of *relative i-cycles* $Z_i(X, A) = Z_i(C(X, A)$ when $i \geq 1$. A element is of the form $z + C_i(A)$ with $\delta(z + C_i(A)) \in C_{i-1}(A)$. By the commutativity of the above diagram, z is then an element of $C_i(X)$ having boundary $\delta(z)$ in $C_{i-1}(A)$. So relative i-cycles are i-cycles of X, or i-chains of X with boundaries in A.

When computing the relative homology groups those i-cycles of X that that are i-boundaries will also become relative i-boundaries. So we only have to consider the relative i-cycles that come from non-trivial i-cycles, and that have boundaries in A. Similarily, if two i-cycles of X differ by an i-boundary, then the corresponding relative i-cycles differ by the corresponding i-boundary. So in calculating $H_i(X, A)$ we only have to consider representative elements of each class in $H_i(X)$.

Problem 7.8. Explain the statement that relative i-boundaries $B_i(X, A) = B_i(C(X, A)) \subset Z_i(X, A)$ are the relative i-cycles of $C_i(X)$ that are homologous to i-chains in A. (What does it mean? Why is it true?)

Problem 7.9. Show that if $H_0(X) \cong \mathbb{Z}$ and A is a (non-empty) subspace of X, then $H_0(X, A)$ is trivial.

Problem 7.10. What can you say in general about $H_0(X, A)$ in terms of $H_0(X)$ (and the distribution of A amoung the path-connected components of X)?

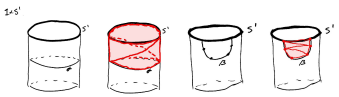
Taking the homology of X relative to one point x_0 can be viewed as analogous to taking the homotopy of X relative to a basepoint. In particular, it should restrict the homology of X to the homology of the component of X containing x_0.

7.5 Example Calculations of Relative Homology

7.5.1 A non-rigourous computation of $H_i(I \times S^1, S^1)$

$H_0(I \times S^1, S^1)$ is trivial as $I \times S$ is path-connected.

The homologically non-trivial 1-cycles in $I \times S^1$ are homologous to a multiples of α (see figure below), and the 1-chains of $i \times S^1$ with boundary in S^1 are generated by 1-chains looking like β in the figure. We give 2-chains showing these to be relative 1-boundaries, by showing them to be homologous to 1-chains in A:



It follows that $H_1(I \times S^1, S^1)$ is trivial.

7.5.2 Computation of $H_i^{\Delta}(S^2, S^1)$: Part 1

Lets compute the simplicial version of the relative homology groups $H_i(S^2, S^1)$ for $i = 0, 1, 2$. These are defined exactly as the relative singular homology groups are, only our initial chain complexes $C(X)$ and $C(A)$ are those defined from a given Δ-complex structure on X.

Let $\Delta(X)$ be the Δ-complex structure on S^2 consisting of two 2-simplices with the same boundary. And let S^1 be this boundary. (This is the Δ-complex structure pictured at the end of Subsection 4.1.)

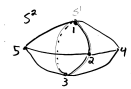
$H_0(X, A)$ is trivial, as $Z_0(X, A) = C_0(X, A) = C_0(X)/C_0(A) \cong 0$ as all 0-simplices of X are 0-simplices of A.

Similarily $C_1(X, A)$ is trivial, and so $Z_1(X, A)$ and so $H_1(X, A)$ are.

H_2 is not trivial. $C_3(X, A)$ is trivial and so $B_2(X, A)$ is, but $C_2(X, A) = C_2(X)/C_2(A) = \mathbb{Z}^2/0 \cong \mathbb{Z}^2$. As $C_1(X, A) = 0$, $Z_2(X, A) = C_2(X, A)$, and so $H_2(X, A) = Z_2/B_2 = \mathbb{Z}^2/0 \cong \mathbb{Z}^2$.

So we have $H_2(X, A) \cong Z^2$ and the other relative homology groups are trivial.

7.5.3 Computation of $H_i^{\Delta}(S^2, S^1)$: Part 2



We do it again with a different Δ-complex structure. This one: The result should be the same, but the process is informative.

Again C_3 and so B_2 are trivial. $Z_2(S^2, S^1)$ is the module of of 2-chains of X with boundary in A. One can show that that it is generated by the chains $[1, 2, 4] + [2, 3, 4] - [1, 3, 4]$ and $[1, 2, 5] + [2, 3, 5] - [1, 3, 5]$. And so $H_2 = Z_2/B_2 = \mathbb{Z}^2/0 \cong Z^2$.

$B_1(S^2, S^1)$ is the module of 1-cycles in S^2 homologous to a cycle in S^1. One can show that all cycles are such, so $B_1(S^2, S^1) \cong C_1(S^2)$. So as $B_1(S^2, S^1) \subset Z_1(S^2, S^1)$ which is a quotient of $C_1(S^2, S^1)$ which is a quotient of $C_1(S^2)$, we have that all of these groups are isomorphic, and so $H_1 = Z_1/B_1$ is trivial.

Now $Z_0 = C_0(X, A) = C_0(X)/C_0(A) = \mathbb{Z}^5/\mathbb{Z}^3 \cong \mathbb{Z}^2$ with generators [4] and [5]. These are both homologous to points in A, so are in B_0. Thus H_0 is trivial.

7.5.4 Non-rigourous Computation of $H_i(S^2, S^1)$

Let's compute the relative homology one more time, this time the singular homology,

For H_0, we have that $Z_0(S^2, S^1) = C_0(S^2, S^1) \cong C_0(S^2)/C_0(S^1)$ is pretty big: $\oplus_{s \in S^2 \setminus S^1} \mathbb{Z}$. But every generator is homologous to a point in A, so $B_0 = Z_0$ and H_0 is trivial.

For H_1, $Z_1(S^2, S^1)$ is generated by 1-simplices in S^2 with endpoints in A. These are all homologous to 1-chains in S^1, so H_1 is trivial. (Showing this is non-trivial, but you can define a 2-simplex with one boundary edge the 1-simplex in S^2 and the other two boundary edges making a path in S^1 between th endpoints of the first edge.)

For H_2, observe that $H_2(X) \cong \mathbb{Z}$ and choose a representative α of the generator. Consider two simplices n and s in S^2 that are homeomorphic maps onto the northern and southern hemispheres respectively, with boundaries mapping to S^1. Try to convince yourself of the following (which are quite difficult to prove).

- Any such n (resp s) are homologous.
- α is homologous to $n + s$.
- Any 2-chain with boundary in S^1 is homologous to something generated by n and s.

It follows that $H_2(S^2, S^1) \cong \mathbb{Z}$.

7.6 Long Exact Sequence of Homology

In this section we will relate the homology of X relative to A to the homologies of X and A, and in the next section, will show that it is isomorphic to the homology of the quotient space X/A. This will provide a useful tool in computing the homology of a space.

Viewing the $C_i(A)$ as a submodule of $C_i(X)$ via the injection, the sequence

$$0 \to C_i(A) \xrightarrow{\text{inj}} C_i(X) \xrightarrow{\phi} C_i(X, A) \to 0$$

where ϕ is the canonical quotient map, is exact. Indeed, by the commutativity of the diagrams above

$$0 \to C(A) \xrightarrow{\text{inj}} C(X) \xrightarrow{\phi} C(X, A) \to 0$$

is a short exact sequence of chain maps. It thus induces the long exact sequence on homology.

Theorem 7.4. *The long sequence*

$$\cdots H_{i+1}(X,A) \xrightarrow{\delta} H_i(A) \to H_i(X) \to H_i(X,A) \xrightarrow{\delta} H_{i-1}(A) \to \cdots$$

where the connecting map is well defined by $\delta = \mathrm{inj} \circ \delta \circ \phi^{-1}$ *which takes* $[z]$*, the class of a cycle* z *in* X *with boundary in* A *to the class* $[\delta(z)]$ *if its boundary in* A*, is exact.* $\square$

Corollary 7.5. *If* $H_{i+1}(X,A)$ *and* $H_i(X,A)$ *are trivial, then* $H_i(A) \cong H_i(X)$.

Observe that the long sequence ends with

$$H_0(X) \to H_0(X,A) \to 0.$$

Exactness here means that $\phi_* : H_0(X) \to H_0(X,A)$ is surjective. Compare this to what you saw in Problem 7.10.

Example 7.6. We computed that $H_0(I \times S^1, S^1) \cong 0$. It follows from the long exact sequence that $\mathrm{inj}_* : H_1(S^1) \to H_1(I \times S^1)$ is surjective, so $H_1(I \times S^1)$ is some quotient of $\mathbb{Z}$. We used this in making the assertion in 7.5.1 that the non-trivial 1-cycles were generated (upto homology) by α. (We used this, in turn, to calculate $H_1(I \times S^1, S^1)$.

Problem 7.11. Show that if A is empty, then $H_i(X) \cong H_i(X,A)$ for all $i \geq 0$.

Problem 7.12. For x_0 a single point in X show that $H_i(X) \cong H_i(X, x_0)$ for all $i > 0$.

Hint:

Only $i = 1$ is a little tricky. For this you should think about Problem 7.10.

7.7 Induced Maps on Relative Homology

A *pair* is a pair (X,A) of spaces such that A is a subspace of X. A continuous function $f : X \to Y$ is called a *pair map* of the pairs (X,A) and (Y,B) if $f(A) \subset B$.

Such pair map clearly commutes with the inclusions $\mathrm{inj} : A \to X$ and $\mathrm{inj} : B \to Y$ as the maps $f \circ \mathrm{inj}$ and $\mathrm{inj} \circ f$ of $A \to Y$ are both simply the restriction of f to A. So extending homomorphically to the groups of i-chains on the respective spaces, the induce the commuting diagram on the left.

$$\begin{tikzcd} C_i(A) \arrow[r, "\mathrm{inj}_\sharp"] \arrow[d, "f_\sharp"] & C_i(X) \arrow[d, "f_\sharp"] \\ C_i(B) \arrow[r, "\mathrm{inj}_\sharp"] & C_i(Y) \end{tikzcd}$$

By Problem 6.2 this extends to the commuting:

$$\begin{tikzcd} C_i(A) \arrow[r, "\mathrm{inj}_\sharp"] \arrow[d, "f_\sharp"] & C_i(X) \arrow[r, "\phi"] \arrow[d, "f_\sharp"] & C_i(X)/C_i(A) \arrow[d, "f_*"] \\ C_i(B) \arrow[r, "\mathrm{inj}_\sharp"] & C_i(Y) \arrow[r, "\phi"] & C_i(Y)/C_i(B) \end{tikzcd}$$

We saw in Subsection 7.2 that the $f_\sharp$ in the middle column commutes with the boundary map δ. The induced map f_* on the factor groups thus commutes with δ, as it does on representatives, and so is a chain map between the chain complexes $C(X,A)$ and $C(Y,B)$.

For spaces $B \subset A \subset X$ the inclusion $A \hookrightarrow X$ is a pair map $(A,B) \hookrightarrow (X,B)$ so induces a chain map $C(A)/C(B) \to C(X)/C(B)$ which is also an inclusion as it is the identity on $C(B)$. Similarly the identity $X \to X$ is a pair map $(X,B) \to (X,A)$ so induces a chain map $C(X)/C(B) \to C(X)/C(A)$ which is clearly surjective. So we have the short exact sequence

$$0 \to C(A,B) \to C(X,B) \to C(X,A) \to 0$$

which induces the long exact sequence

$$\cdots H_{i+1}(X,A) \xrightarrow{\delta} H_i(A,B) \to H_i(X,B) \to H_i(X,A) \xrightarrow{\delta} H_{i-1}(A,B) \to \cdots$$

Taking B to be empty, we recover the long exact sequence from Theorem 7.4.

7.8 Excision

The idea of *excision* is that for certain subspaces U of $A \subset X$, the relative homology of the pair (X,A) doesn't change if we remove A, that is

$$H_*(X \setminus U, A \setminus U) \cong H_*(X,A),$$

so we can remove U and hopefully simplify the calculation of the homology.

Theorem 7.7. *Given spaces* $U \subset A \subset X$ *such that the closure of* U *is in* A*, the inclusion* $(X \setminus U, A \setminus U) \hookrightarrow (X,A)$ *induces an isomorphism* $H_i(X \setminus U, A \setminus U) \cong H_i(X,A)$ *for each* $i \geq 0$*. Equivalently, given spaces* $A, B \subset X$ *whose interiors cover* X*, the inclusion* $(B, A \cap B) \hookrightarrow (X,A)$ *induces an isomorphism on relative homology.*

The equivalence of the two statments is seen by taking $B = X - U$.

The first main idea in the proof is that to compute the homology of a space, we only have to consider small simplices: larger ones are constructed from them as chains. So we can assume the simplices are small enough that every one of them is fully within A or B.

For $i \geq 0$ let C_i^U be the subset of $C_i(X)$ generated by i-chains fully contained in A or fully contained in B. It is clear that $C_i(B)/C_i(A \cap B) \cong C_i^U/C_i(A)$ as both are free groups are generated by the classes of i-cycles in B that are not also in A. Thus there is an isomorphism of the relative homology group $H_i(B, A \cap B)$ with the relative homology group $H_i(C^u, C(A))$ (see Problem 7.7.)

To prove the 'Equivalently...' version of the theorem, we thus have to show that $H_i(C^u, C(A))$ is isomorphic to $H_i(X, A) = H_i(C(X), C(A))$.

We do this with the following lemma.

Lemma 7.8. *The inclusion $i : C_i^u \to C_i$ is a chain homotopy equivalence, so the chain complexes C^u and C induce isomorphic homology groups $H_i(C^u) \cong H_i(C) = H_i(X)$. Further, this isomorphism descends to the quotient: $H_i(C^u, C(A)) \cong H_i(C(X), C(A))$*

In the rest of the section we prove Lemma 7.8. The proof takes a couple of steps. We start off by defining cones and the barycentric subdivision of a simplicial complex.

7.8.1 Cones

For an i-simplex $\sigma = [v_0, \dots, v_i]$ in X and a point b, let the *cone* $b \cdot \sigma$ be the $i+1$-simplex

$$b \cdot \sigma = [b, v_0, \dots, v_{i-1}].$$

The operation $\sigma \mapsto b \cdot \sigma$ extends to a homomorphism $C_i(X) \to C_{i+1}(X)$ on i-chains. Further we have

$$\delta(b \cdot \sigma) = \sigma - (b \cdot \delta\sigma), \tag{5}$$

as we can verify on generators by:

$$\delta[b, v_0, \dots, v_i] = [v_0, \dots v_i] - \sum_{j=0}^{i} [b, v_0, \dots, \hat{v_j}, \dots v_i].$$

Problem 7.13. Draw a picture of the cone of a 2-simplex, and interpret the identity on the boundary operator.

7.8.2 Barycentric Subdivision

Let $\Delta^d = [v_0, v_1, \dots v_d]$ be a **linear** d-simplex. Then the *barycentre* b of Δ^d is the, center of gravity of $b = \frac{1}{d+1}\sum_{i=0}^{d} v_i$ of Δ^d. For a d-simplex $\sigma : \Delta^d \to X$ in X, the barycentre b_σ of σ is $\sigma(b)$ where b is the barycentre of Δ^d.

We define the *barycentric subdivision* $S\sigma$ of a d-simplex σ as follows. If $d = 0$, then $S\sigma = \sigma$. Otherwise $S\sigma = b_\sigma \cdot S\delta\sigma$. See page 120 of [Ha] for a picture.

The following is not so hard.

Problem 7.14. Show that $\delta S\sigma = S\delta\sigma$. Illustrate this by computing both when σ is the 2-simplex $[1, 2, 3]$.

Hint:

Induction.

This gives us that $S : C(X) \to C(X)$ is a chain map on the chain $C(X)$ of singular simplical chains modules of X.

7.8.3 A chain homotopy

We show that the barycentric subdivision S is chain homotopic to the identity chain map $\mathrm{id} : C(X) \to C(X)$, so conclude that moving to the barycentric subdivision doesn't affect homology.

We define a chain homotopy h between S and id as follows. Let $h_{-1} : C_{-1} \to C_0$ be the zero map, and for $i \geq 0$ let $h_i : C_i \to C_{i+1}$ be defined on i-simpices σ by

$$h_i\sigma = b_\sigma \cdot (\sigma - h_{i-1}\delta\sigma).$$

We show now that this is a chain homotopy between S and id, we must show that $h\delta + \delta h = \mathrm{id} - S$. Rearranging this to $\delta h = \mathrm{id} - h\delta - S$, we show it by induction on i. That is, we show for every $i \geq 0$ and every i-cycle $\sigma \in C_i(X)$ that $\delta h_i\sigma = \sigma - h_{i-1}\delta\sigma - S\sigma$.

For the induction, we have

$$\begin{aligned}
\delta h_i\sigma &\overset{\text{def of } h}{=} \delta(b_\sigma \cdot (\sigma - h_{i-1}\delta\sigma)) \\
&\overset{(5)}{=} (\sigma - h_{i-1}\delta\sigma) - b_\sigma \cdot (\delta(\sigma - h_{i-1}\delta\sigma)) \\
&= (\sigma - h_{i-1}\delta\sigma) - b_\sigma \cdot (\delta\sigma - \delta h_{i-1}\delta\sigma)) \\
&\overset{\text{induction}}{=} (\sigma - h_{i-1}\delta\sigma) - b_\sigma \cdot (\delta\sigma - (\delta\sigma - \cancel{h_{i-1}\delta\delta\sigma} - S\delta\sigma)) \\
&= \sigma - h_{i-1}\delta\sigma - b_\sigma \cdot S\delta\sigma \\
&\overset{\text{def of } S}{=} \sigma - h_{i-1}\delta\sigma - S\sigma
\end{aligned}$$

Problem 7.15. Show the base case $i = 0$.

7.8.4 Sketch of proof of Lemma 7.8

We have to show that the inclusion $\mathrm{inj} : C^U \to C$ is a chain homotopy equivalence. To do this we must find a chain map $\rho : C \to C^U$ such that $\mathrm{inj} \circ \rho$ and $\rho \circ \mathrm{inj}$ are chain homotopic to the identity.

The map ρ will basically be an iteration of the barycentric subdivision S,(only the number of times we iterate it depends on the simplex we are applying it to).

Since $\mathrm{Int}(A)$ and $\mathrm{Int}(B)$ are an open cover of a compact metric space (which we know they are as they are the image under continuous functions simplices, which are closed real spaces) there is some (Lebesque number) $\varepsilon \geq 0$ such that any open ball of radius ε is contained within $\mathrm{Int}(A)$ or $\mathrm{Int}(B)$.

Observe that if a linear d-simplex has diameter D, then any simplex in its barycentric subdivision has diameter at most $D \cdot \frac{d}{d+1}$. So by iterating the subdivision operation on a linear simplex, we can ensure it is the union of simplices of arbitrarily small diameter.

As a d-simplex σ in X is the continuous image of a linear d-simplex, we can say the same for it: there is some m_σ such that S^{m_σ} (the m_σ time iterated subdivision of σ) is the sum of d-simplices of radius less than ε, so in $\mathrm{Int}(A)$ or $\mathrm{Int}(B)$.

We define $\rho^* : C \to C^U$ by letting $\rho^*(\sigma) = S^{m_\sigma}$ for any singular simplex σ in X. (Now this is not a chain map, due to the inhomogeneity of the m_i, but we will adjust it later.)

We define (what will be our chain homomotopy) $H : C_n \to C_{n+1}$ by $H\sigma = h^{m_\sigma}\sigma$.

If m_σ were the same for σ and everything in its boundary, then it would follow from the fact that h is a homotopy between id and S, that H is a homotopy between id and ρ^*:

$$H\delta\sigma + \delta H\sigma = h^{m_\sigma}\delta\sigma + \delta h^{m_\sigma}\sigma = \sigma - S^{m_\sigma}\sigma = (\mathrm{id} - \rho^*)\sigma$$

But in reality $H\delta\sigma$ is different from $h^{m_\sigma}\delta\sigma$, so to make this equation work, we let $\rho\sigma = \rho^*\sigma - h^{m_\sigma}(\delta\sigma) + H(\delta\sigma)$. This gives us that H **is** a homotopy between id and ρ:

$$H\delta\sigma + \delta H\sigma = \sigma - [S^{m_\sigma} - h^{m_\sigma}(\delta\sigma) + H(\delta\sigma)] = (\mathrm{id} - \rho)\sigma$$

Now we have just to verify that ρ is indeed a chain map from C C^U. To see that ρ maps to C^U we note that m_τ is only smaller than m_σ for some $\tau \in \delta\sigma$ when $S^{m_\tau}\tau$ is in C^U anyways. To see that it is a chain map we rearrange identity $H\delta\sigma + \delta H\sigma = \sigma - \rho\sigma$, as $\rho\sigma = \sigma - H\delta\sigma - \delta H\sigma$ Then $\delta\rho(\sigma) = \delta(\sigma) - \delta H\delta\sigma = \rho(\delta\sigma)$, as needed.

This completes the proof of the main statement of the lemma, that $\mathrm{inj}\, C^U \to C$ is a chain homotopy equivalence. Thus we have an isomorphism of homology $H_i(C^U) \cong H_i(C)$. To see that the isomorphism descends to the relative homology groups $H_i(C^u/C(A)) \cong H_i(C/C(A))$, observe that the maps H and ρ restrict on $C(A)$ to maps on $C(A)$. So they induce quotient maps on $C/C(A)$ and $C^U/C(A)$ which satisfy the same identities. In particular this give an chain homotopy equivalence $\mathrm{inj}^*\, C^U/C(A) \to C/C(A)$.

7.9 Cellular Homology

7.10 Mayer-Vietoris Sequence

7.11 Simplicial vs. Singular Homology

8 More Homological Algebra

8.1 Tensor Products

8.2 Universal Coefficient Theorem

8.3 The Kunneth Formula

References

[Ha] Allan Hatcher, 'Algebraic Topology (2002).' 'http://www.math.cornell.edu/ hatcher/